SingleRAN

X2 and S1 Self-Management in NSA Networking Feature Parameter Description

Issue Draft B

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Added a statement that the **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NetworkPrfmOptSwitch*** parameter can be selected only in RAN sharing with common carrier scenarios. For details, see **4.7.2.1 Prerequisite Functions**.

Revised other descriptions in this document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 07 (2025-12-14).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for "prohibition of multi-operator sharing during X2-U self-setup". For details, see: • **4.1.1 Overview** • **4.8.1.1 Data Preparation** • **4.8.1.2 Using MML Commands**	Modified parameter on the NR side: Added the **X2_UP_SELF_SETUP_NO_MOCN_SW** option to the **gNBInterfaceParam**.***InterfaceSelfCfgOptSw*** parameter.	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added support for X2 link setup for inter-base-station coordination services. For details, see: • **4.1.2.2 X2 Self-Setup with Peers Automatically Configured** • **4.3 X2 Self-Removal** • **4.6.2 Impacts**	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added statements that the BTS5929R does not support X2 self-management, direct IPsec for the X2 interface, and S1-U self-management. For details, see: • **4.7.3 Hardware** • **5.4.3 Hardware** • **6.3.3 Hardware**	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Editorial Changes

Revised descriptions in the document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-131121	LTE and NR X2 Interface Self-Configuration (LTE FDD)	• **4 X2 Self-Management** • **5 S1-U Self-Management** • **6 Direct IPsec for X2 Interface**
LTE TDD	MRFD-131131	LTE and NR X2 Interface Self-Configuration (LTE TDD)	
NR	MRFD-131161	LTE and NR X2 Interface Self-Configuration (NR)	

2.4 Differences

Table 2-1 Differences between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
X2 self-management	None	**4 X2 Self-Management**
S1-U self-management	None	**5 S1-U Self-Management**
Direct IPsec for X2 interface	None	**6 Direct IPsec for X2 Interface**

Table 2-2 Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
X2 self-management	None	**4 X2 Self-Management**
S1-U self-management	None	**5 S1-U Self-Management**
Direct IPsec for X2 interface	None	**6 Direct IPsec for X2 Interface**

Table 2-3 Differences between NSA and SA

Function Name	Difference	Chapter/Section
X2 self-management	Supported only in NSA networking	**4 X2 Self-Management**
S1-U self-management	Supported only in NSA networking	**5 S1-U Self-Management**
Direct IPsec for X2 interface	Supported only in NSA networking	**6 Direct IPsec for X2 Interface**

Table 2-4 Differences between high frequency bands and low frequency bands

Function Name	Difference	Chapter/Section
X2 self-management	None	**4 X2 Self-Management**
S1-U self-management	None	**5 S1-U Self-Management**
Direct IPsec for X2 interface	None	**6 Direct IPsec for X2 Interface**

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

3GPP Release 15 has introduced support for E-UTRA-NR dual connectivity (EN-DC) architecture. **Figure 3-1** and **Figure 3-2** illustrate the logical architecture of EN-DC with base stations in integrated deployment mode.

Figure 3-1 Logical architecture of the EN-DC interfaces (Option 3x networking)

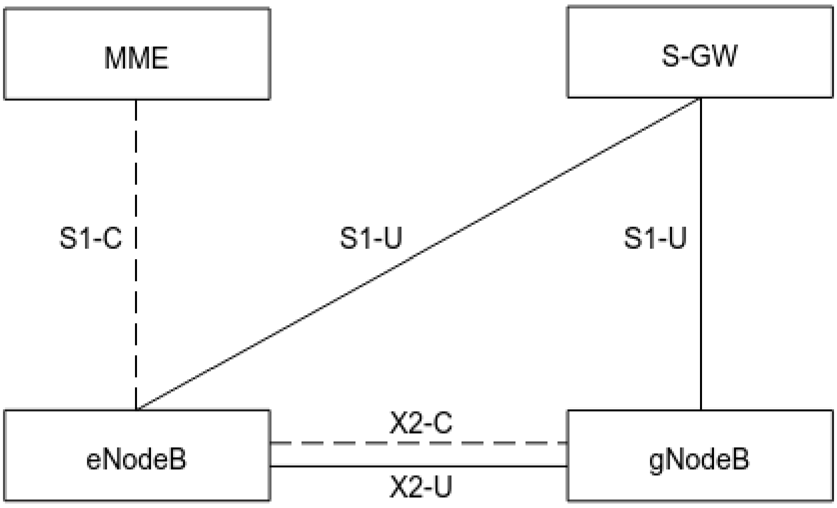
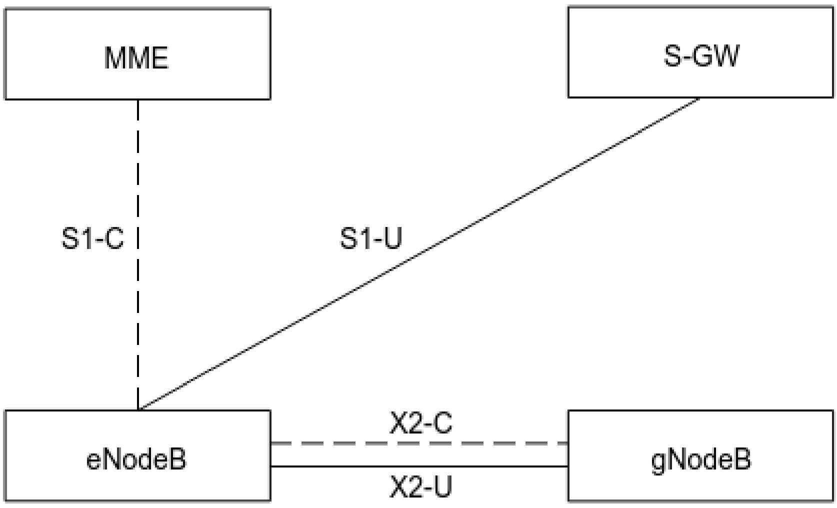


Figure 3-2 Logical architecture of the EN-DC interfaces (Option 3 networking)



In EN-DC:

- X2 interface is the logical interface between the eNodeB and the gNodeB. The X2 interface includes the X2 interface on the control plane (X2-C) and X2 interface on the user plane (X2-U). X2-C and X2-U forward control- and user-plane data between an eNodeB and a gNodeB, respectively. The X2 control plane is based on the Stream Control Transmission Protocol (SCTP) and the user plane is based on the GPRS Tunneling Protocol-User Plane (GTP-U).

  NOTE

  For details about how to configure X2 interfaces on eNodeBs, see *S1 and X2 Self-Management* in eRAN feature documentation.

  For details about the Option 3x and Option 3 networking, see *EPC-based NSA Basics*.

- S1-U, the user-plane interface between the gNodeB and S-GW, uses GTP-U to carry user-plane data.

  NOTE

  If DC services are initiated for UEs when Option 3x architecture is used for data splitting, an S1-U interface must be configured on the gNodeB. If DC services are initiated for UEs when Option 3 networking is used for data splitting, S1 interface is not required on the gNodeB. (An S1 interface is configured on the eNodeB.) For details about how to configure S1 interfaces on eNodeBs, see *S1 and X2 Self-Management* in eRAN feature documentation. To simplify management, self-management is implemented for the X2 and S1-U interfaces in EN-DC.

- The X2 interface supports self-setup, self-update, and self-removal.
- The S1-U interface supports self-setup and self-removal.

4 X2 Self-Management

4.1 X2 Self-Setup

4.1.1 Overview

In EN-DC, the X2 interface can be configured only in endpoint mode for an eNodeB or a gNodeB.

NOTE

An endpoint mode is a type of transmission link configuration mode based on endpoints. When configuring a transmission link, you need to configure endpoint information including the control-plane host and peer and user-plane host and peer. NR does not support the simplified endpoint mode. For details about the simplified endpoint mode of LTE, see *S1 and X2 Self-Management*.

Since the LTE side of an LTE and NR co-MPT base station on an NSA network does not support the simplified endpoint mode, the mode must be disabled on the LTE side during the reconstruction from an eNodeB to such a co-MPT base station.

The user-plane and control-plane hosts of an X2 interface must be configured manually. For details, see **4.8.1.2 Using MML Commands**. The user-plane and control-plane peers can be configured manually or automatically, which must be consistent between the eNodeB and gNodeB.

- Peers manually configured

  In this scenario, **SCTPPEER** and **USERPLANEPEER** MOs must be configured for the peers manually. If an eNodeB or a gNodeB has multiple X2 interfaces, multiple peer configurations must be manually added.

- Peers automatically configured

  In this scenario, the peers do not need to be configured manually. When EN-DC services are initiated for UEs on the eNodeB side, if the eNodeB detects that the X2 link does not exist or is faulty, it exchanges X2 transport layer configuration data with the gNodeB through the MAE. Through this procedure, the eNodeB obtains the IP address of the gNodeB X2 interface. The local port number, specified by **SCTPHOST**.***PN (5G gNodeB, LTE eNodeB)***, used by the SCTP link of the eNodeB must be the same as that of the gNodeB. Otherwise, SCTP link negotiation will fail. In the current version, DC services cannot be initiated for UEs on the gNodeB side.

In an LTE and NR co-MPT base station, when X2 links are set up between the eNodeB and its neighboring eNodeBs as well as between the gNodeB and its neighboring eNodeBs, the local SCTP port number, specified by **SCTPHOST**.***PN (5G gNodeB, LTE eNodeB)***, used by the eNodeB in the co-MPT base station must be the same as that used by the gNodeB.

For X2 interface requirements in NSA multi-operator sharing scenarios, see *Multi-Operator Sharing* in *5G RAN Feature Documentation*.

The X2 interfaces on the eNodeB in EN-DC and the inter-eNodeB X2 interfaces share X2 interface resources. If the self-setup function with peers automatically configured is enabled but self-removal is disabled for inter-eNodeB X2 interfaces, a large number of redundant X2 links may exist, leading to a resource waste. To address this issue, it is recommended that X2 self-removal based on link faults and X2 self-removal based on X2 usage be enabled for inter-eNodeB X2 interfaces in NSA networking. If the number of established X2 interfaces has reached the maximum value, EN-DC X2 interfaces cannot be set up. To address this issue, it is recommended that inter-eNodeB or LTE-NR inter-RAT X2 self-removal based on dynamic sharing of X2 specifications be enabled. For details about inter-eNodeB X2 self-removal, see *S1 and X2 Self-Management* in *eRAN Feature Documentation*.

NOTE

- Maximum number of X2-C interfaces supported on the LTE and NR sides:
  - LTE: UMPTb: 256; UMPTe: 384; UMPTga: 512; UMPTg/UMPTha: 768
  - NR: UMPTe/UMPTb/UMPTga: 384; UMPTg/UMPTha: 768
- Both X2 over S1 and X2 over OSS are supported for X2 self-setup with peers automatically configured between eNodeBs. However, only X2 over OSS is supported for X2 self-setup with peers automatically configured between eNodeBs and gNodeBs. For details about X2 self-setup with peers automatically configured between eNodeBs, see *S1 and X2 Self-Management* in *eRAN Feature Documentation*.
- X2 self-setup on only the eNodeB or gNodeB is not supported in NSA networking.
- For an LTE and NR co-MPT base station, the local IP address of the X2 interface on the eNodeB must differ from that of the X2 interface on the gNodeB. In NSA and SA hybrid networking, the local IP address of the Xn interface and that of the X2 interface on the gNodeB can be identical. When the IP addresses of the control-plane hosts of the Xn and X2 interfaces are identical, the local SCTP port numbers, specified by **SCTPHOST.*PN (5G gNodeB, LTE eNodeB)***, of the Xn and X2 interfaces must be different.
- In RAN sharing with common carrier mode, to separate X2-U link self-setup/self-removal procedures for different operators, the **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NetworkPrfmOptSwitch*** parameter on the eNodeB side and the **X2SON_X2U_AUTO_DEL_OPT_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter and the **X2_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam**.***InterfaceSelfCfgOptSw*** parameter on the gNodeB side need to be selected at the same time.

The source base station starts a 10-minute timer after initiating an X2 self-setup to a target base station. A second X2 self-setup to the target base station will not be initiated before the timer expires. This prevents frequent X2 self-setups between them.

4.1.2 IPv4/IPv6 Single-Stack Transmission

4.1.2.1 X2 Self-Setup with Peers Manually Configured

An X2 interface is called X2-C when it is used for the control plane and is called X2-U for the user plane. **Figure 4-1** shows the relationships between MOs involved in X2 self-setup with peers manually configured.

Figure 4-1 Relationships between MOs involved in X2 self-setup with peers manually configured

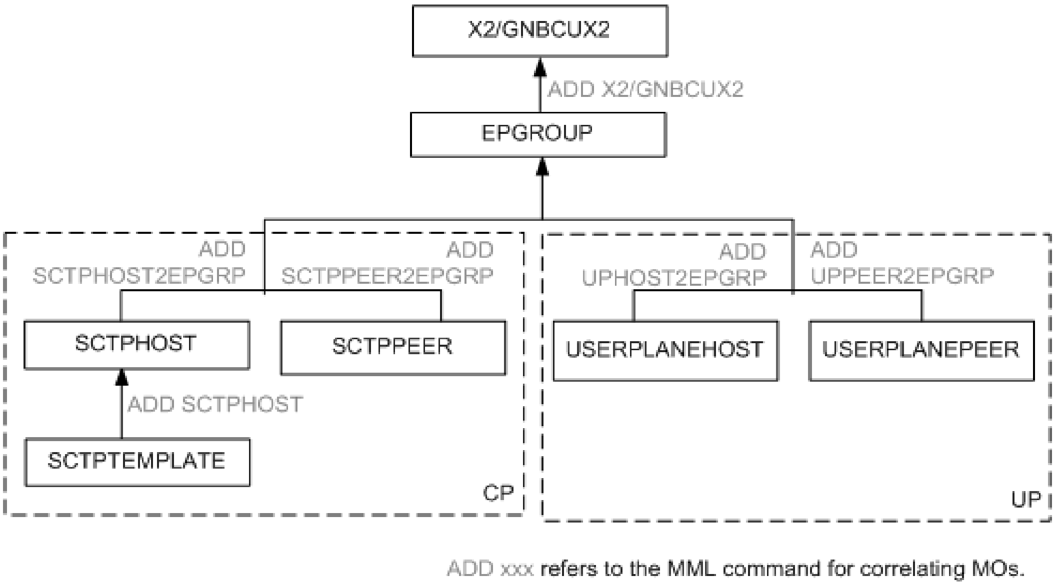


Table 4-1 describes the relationships between the MOs shown in **Figure 4-1**.

Table 4-1 Setting notes for the MOs involved in X2 self-setup with peers manually configured

Object	MO at the Local End	MO at the Peer End	Description
Control plane	**SCTPHOST** **NOTE** The **SCTPHOST**.***SCTPTEMPLATEID (LTE eNodeB, 5G gNodeB)*** and **SCTPTEMPLATE**.***SCTPTEMPLATEID (LTE eNodeB, 5G gNodeB)*** parameter values must be the same. In this way, the attribute parameters of the control-plane links are configured consistently with those in the SCTP parameter template.	**SCTPPEER**	These MOs are used to set parameters such as the local and peer IP addresses and the port numbers of a control-plane link. ● The **SCTPHOST**.***PN (LTE eNodeB, 5G gNodeB)*** and **SCTPPEER**.***PN (LTE eNodeB, 5G gNodeB)*** parameters must be set to the same value. The recommended value is **36422**. For details, see chapter 7 "Transport layer" in 3GPP TS 36.422 V15.1.0. ● The first local IP address (specified by the **SCTPHOST**.***SIGIP1V4 (LTE eNodeB, 5G gNodeB)*** parameter) and the first peer IP address (specified by the **SCTPPEER**.***SIGIP1V4 (LTE eNodeB, 5G gNodeB)*** parameter) are used as the local and peer IP addresses of the SCTP link, respectively. The eNodeB or gNodeB automatically sets up a multihoming SCTP link if a second IP address is configured in the **SCTPHOST** or **SCTPPEER** MO.

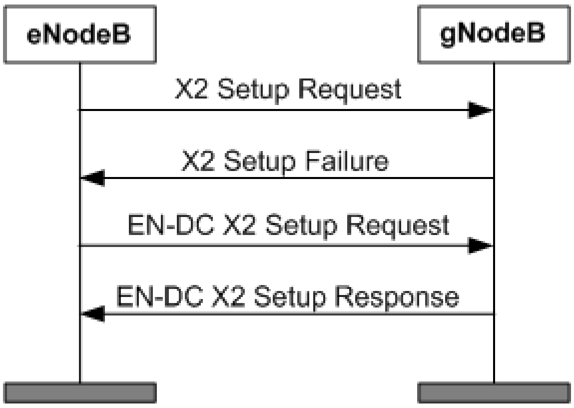
Object	MO at the Local End	MO at the Peer End	Description
User plane	**USERPLANEHOST**	**USERPLANEPEER**	These MOs are used to set parameters such as the local and peer IP addresses of a user-plane link. The eNodeB or gNodeB automatically sets up a transmission link for the X2-U interface using the local X2 user-plane IP address (specified by the **USERPLANEHOST**.***LOCIPV4 (LTE eNodeB, 5G gNodeB)*** parameter) and the peer gNodeB or eNodeB IP address (specified by the **USERPLANEPEER**.***PEERIPV4 (LTE eNodeB, 5G gNodeB)*** parameter). If the **USERPLANEPEER** MO is configured for the eNodeB, such configuration is not required for the gNodeB and self-setup can be directly performed.
Endpoint group	**EPGROUP**	**EPGROUP**	When adding the control-plane host and peer as well as the user-plane host and peer to endpoint groups: ● The control-plane host and peer of an X2 interface must be added to the same endpoint group. ● The user-plane host and peer of an X2 interface must be added to the same endpoint group.
X2 interface	**X2**	**X2**	This MO is used to add an X2 object on the eNodeB and bind it with the endpoint group.
X2 interface	**gNBCUX2**	**gNBCUX2**	This MO is used to add an X2 object on the gNodeB and bind it with the endpoint group.

After the preceding MOs are configured, the eNodeB automatically initiates X2 self-setup.

- If the **SCTPPEER** MO is automatically configured, the local eNodeB can identify that the peer NE is a gNodeB. The eNodeB then sends an EN-DC X2 Setup Request message for X2 self-setup.

- If the **SCTPPEER** MO is manually configured, the local eNodeB cannot determine whether the peer NE is an eNodeB or a gNodeB. In this case, the local eNodeB initiates an X2 Setup Request message, as shown in **Figure 4-2**. After receiving this message, the peer gNodeB responds with an X2 Setup Failure message with the cause value of "scg-mobility." The eNodeB then determines that the peer NE is a gNodeB, and sends an EN-DC X2 Setup Request message. An X2 interface is then set up based on the procedure defined in 3GPP TS 36.423 R15.

Figure 4-2 Procedure of X2 self-setup with peers manually configured



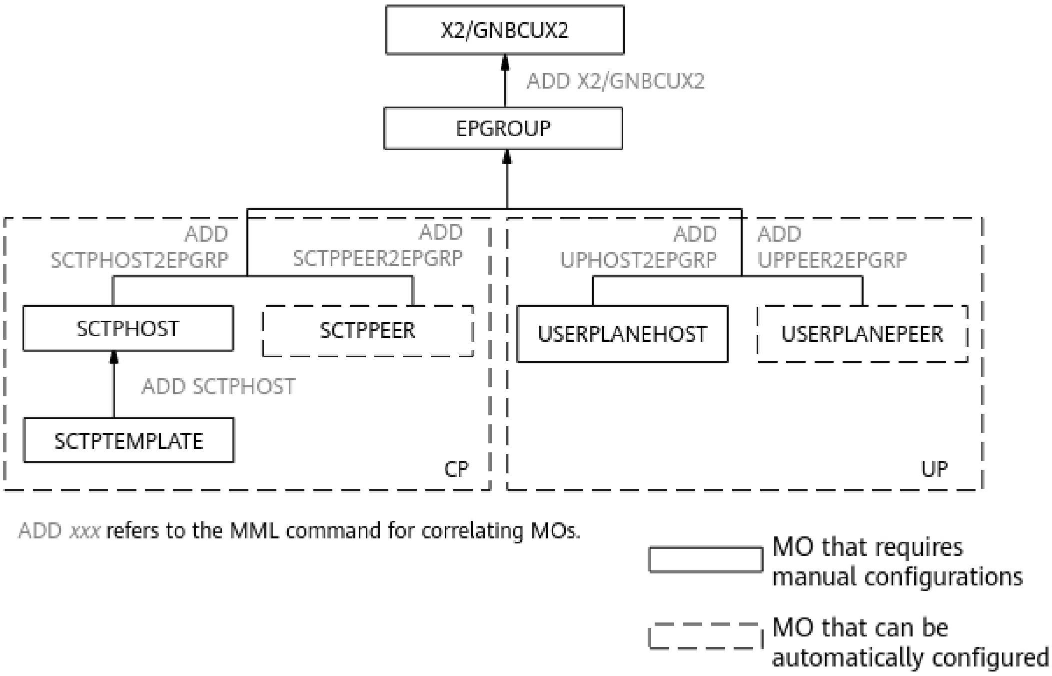
NOTE

- Assume that the manually configured X2-C link is functioning properly, but the X2-U link is yet to be configured. If an NSA UE initiates a DC service and the MeNB needs to add an SgNB, the eNodeB sends an SgNB Addition Request message containing the local X2-U address to the gNodeB, and the gNodeB sends an SgNB Addition Request Acknowledge message containing the local X2-U address to the eNodeB. When both the gNodeB and eNodeB obtain the X2-U address of the peer base station, they both initiate an X2-U self-setup procedure. For details about the signaling procedure, see *NSA Mobility Management*.
- NR cells must be activated before X2 self-setup. If no NR cell is activated, the gNodeB sends an EN-DC X2 Setup Failure message to the eNodeB during X2 self-setup. As a result, the X2 link is faulty.
- In scenarios where direct IPsec is or is not used for the user-plane host, if the control plane is normal, the user-plane IP address of the peer base station can be carried in a control-plane message. In this case, the **USERPLANEPEER** MO can be automatically created.

4.1.2.2 X2 Self-Setup with Peers Automatically Configured

X2 self-setup with peers automatically configured, as shown in **Figure 4-3**, is recommended because it is free from manual operations, reducing costs on network planning, optimization, and O&M.

Figure 4-3 Relationships between MOs involved in X2 self-setup with peers automatically configured



X2 self-setup with peers automatically configured is triggered when the eNodeB establishes DC services for UEs.

To enable X2 self-setup with peers automatically configured on the eNodeB, both the **LTE_NR_X2_SON_SETUP_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter on the eNodeB and the **X2SON_SETUP_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter on the gNodeB must be selected. X2 self-setup is triggered when the eNodeB initiates an SgNB addition procedure.

For details about the inter-MeNB handover when an X2-U peer is automatically configured, see *NSA Mobility Management*.

NOTE

For X2 self-setup with peers automatically configured on an eNodeB or a gNodeB, two user-plane peers (**USERPLANEPEER** MOs) can be automatically created at the same time. If there are self-setup requests from more than two user-plane peers at the same time, excess requests will be rejected, which may result in UE access failures.

In addition to DC services for UEs, X2 links are also set up proactively when inter-base-station coordination services are required. For examples of inter-base-station coordination services, see descriptions of support for LTE and NR intelligent carrier shutdown in inter-base-station SA networking in *Multi-RAT Coordinated Energy Saving*.

4.1.2.2.1 OSS-based Automatic Peer Configuration for the eNodeB and gNodeB

Automatic Peer Configuration for the eNodeB and gNodeB

Figure 4-4 shows the X2 self-setup procedure when the eNodeB and gNodeB are managed by the same OSS.

In this mode, the eNodeB, gNodeB, and MAE must meet the following conditions:

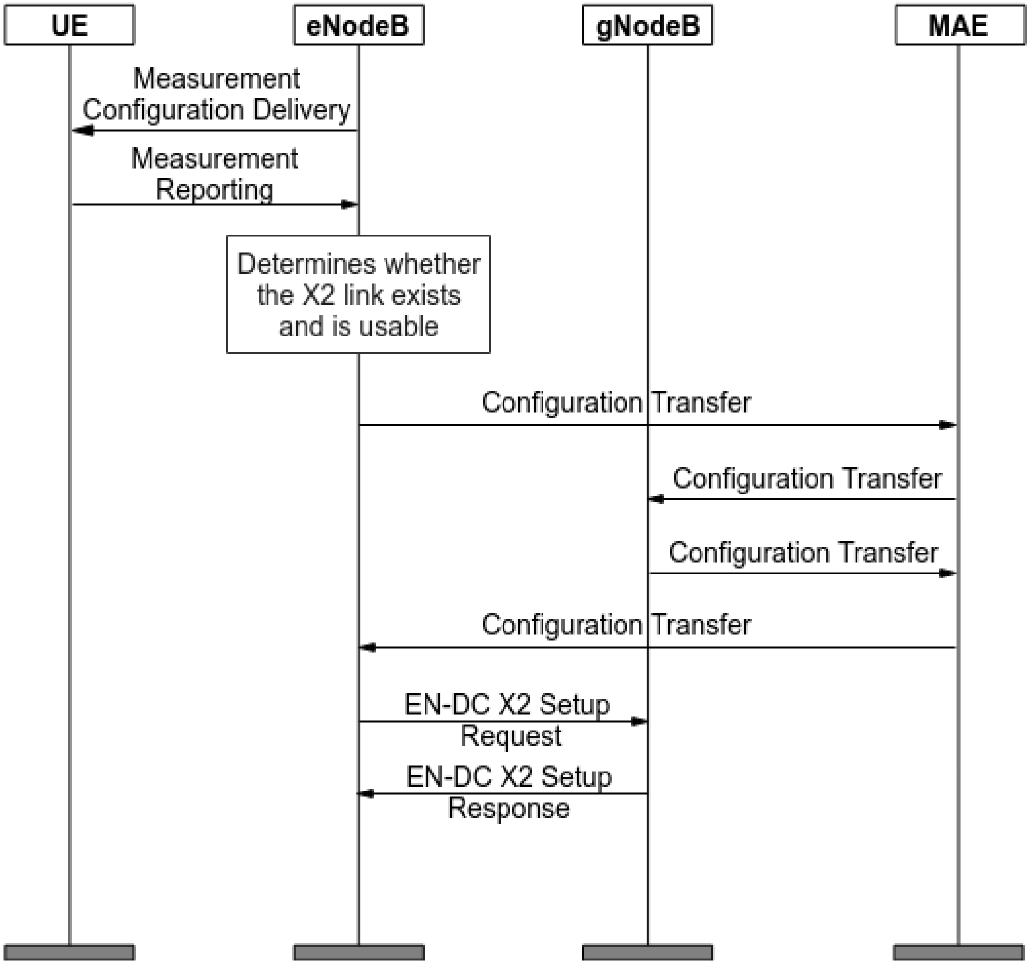
- For the eNodeB
  - The following MOs have been configured: **SCTPHOST**, **USERPLANEHOST**, **X2**, **EPGROUP**, and **SCTPTEMPLATE**.
  - Neighbor relationships with NR cells have been configured.

    NOTE

    If the **X2_SON_SETUP_NO_NR_NRT_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter is selected, neighbor relationships with NR cells do not need to be configured.
- For the gNodeB

  The following MOs have been configured: **SCTPHOST**, **USERPLANEHOST**, **gNBCUX2**, **EPGROUP**, and **SCTPTEMPLATE**.
- For the MAE
  - The CMServer service has been enabled. Otherwise, the MAE cannot respond to the X2 self-setup request from the eNodeB. For details about CMServer, see **Description** > **Function Description** > **MAE-Access Function Description** > **MAE Function Description About Software Management** > **Processes and Services** in *iMaster MAE Product Documentation (EulerOS, TaiShan)*.
  - The eNodeB and gNodeB are managed by the same MAE.

Figure 4-4 X2 self-setup with peers automatically configured for the eNodeB and gNodeB



The procedure is as follows:

1. To set up DC for a UE, the source eNodeB delivers measurement configurations for neighboring NR cells. For details about how the base station determines whether to set up DC for a UE, see *EPC-based NSA Basics*.
2. The UE performs measurements as instructed after receiving the measurement configurations, and sends measurement reports about neighboring NR cells to the source eNodeB.
3. The source eNodeB determines whether an X2 link with the gNodeB serving the reported neighboring NR cells exists and is usable.
   – If the X2 link exists and is operational, no X2 self-setup procedure will be triggered.
   – If the X2 link is faulty or does not exist, an X2 self-setup procedure will be triggered. The procedure goes to **4**.
4. The source eNodeB identifies the physical cell identifier (PCI) in the UE measurement report and determines the gNodeB ID corresponding to the PCI based on neighboring NR cells configured on the LTE side. Then, the source eNodeB sends a Configuration Transfer message to the MAE. The message contains the information about the X2 interface of the source eNodeB, including the X2-C IP address, X2-U IP address, and operator ID.

5. After receiving the message, the MAE forwards it to the target gNodeB according to the target gNodeB ID.

   NOTE

   If the MAE fails to find the target gNodeB, the X2 self-setup procedure is terminated as a failure.

6. Upon receiving the message, the target gNodeB sends a Configuration Transfer message carrying its control- and user-plane IP addresses to the MAE. In addition, the target gNodeB automatically generates related MOs to set up transmission links for the control plane and user plane.
   a. The target gNodeB automatically generates the **USERPLANEPEER** and **SCTPPEER** MOs based on eNodeB's control- and user-plane IP addresses included in this message. The target gNodeB then automatically adds the generated MOs to the **EPGROUP** MO.
   b. The target gNodeB automatically generates the **gNBCUX2Interface** MO based on manually configured MOs and automatically created MOs. Manually configured MOs include **SCTPHOST**, **USERPLANEHOST**, **gNBCUX2**, **EPGROUP**, and **SCTPTEMPLATE**. Automatically created MOs include **SCTPPEER** and **USERPLANEPEER** generated in **a**.
7. The MAE forwards the message to the source eNodeB.

   Upon receiving the message, the source eNodeB automatically generates related MOs to set up transmission links for the control plane and user plane.
   a. The source eNodeB automatically generates the **USERPLANEPEER** and **SCTPPEER** MOs based on the control- and user-plane IP addresses of the target end included in this message. The source eNodeB then automatically adds the generated MOs to the **EPGROUP** MO.
   b. The source eNodeB automatically generates the **X2Interface** MO based on manually configured MOs and automatically created MOs. Manually configured MOs include **SCTPHOST**, **USERPLANEHOST**, **X2**, **EPGROUP**, and **SCTPTEMPLATE**. Automatically created MOs include **SCTPPEER** and **USERPLANEPEER** generated in **a**.

   NOTE

   If the MAE fails to find the source eNodeB, the X2 self-setup procedure fails.

8. The source eNodeB then sends an EN-DC X2 Setup Request message to the target gNodeB.
9. After receiving the message, the target gNodeB sends an EN-DC X2 Setup Response message to the source eNodeB to complete X2 self-setup.

4.1.2.2.2 Automatic Peer Configuration Between gNodeBs

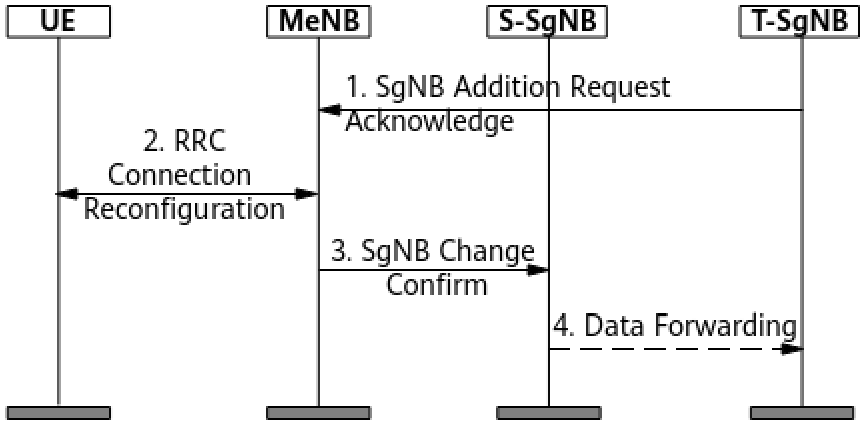
In NSA networking, to prevent packet loss during an inter-gNodeB handover, data sent to the source gNodeB needs to be forwarded to the target gNodeB. This process is known as "data forwarding". During this process, a unidirectional X2-U link from the source gNodeB to the target gNodeB is automatically set up to transfer data from the source gNodeB to the target gNodeB. **Figure 4-5** illustrates the data forwarding process during the inter-gNodeB handover, including the setup of the X2-U link between gNodeBs.

To forward data between gNodeBs through the X2-U link, the following conditions must be met:

- An X2-U transmission link has been set up.
- The following MOs have been configured: **USERPLANEHOST**, **gNBCUX2**, and **EPGROUP**.

The automatic peer configuration between gNodeBs in non-direct IPsec scenarios differs from that in direct IPsec scenarios. The processes are shown in **Figure 4-5** and **Figure 4-6**, respectively. In the figures, the MeNB is a master base station and the SgNB is a secondary base station.

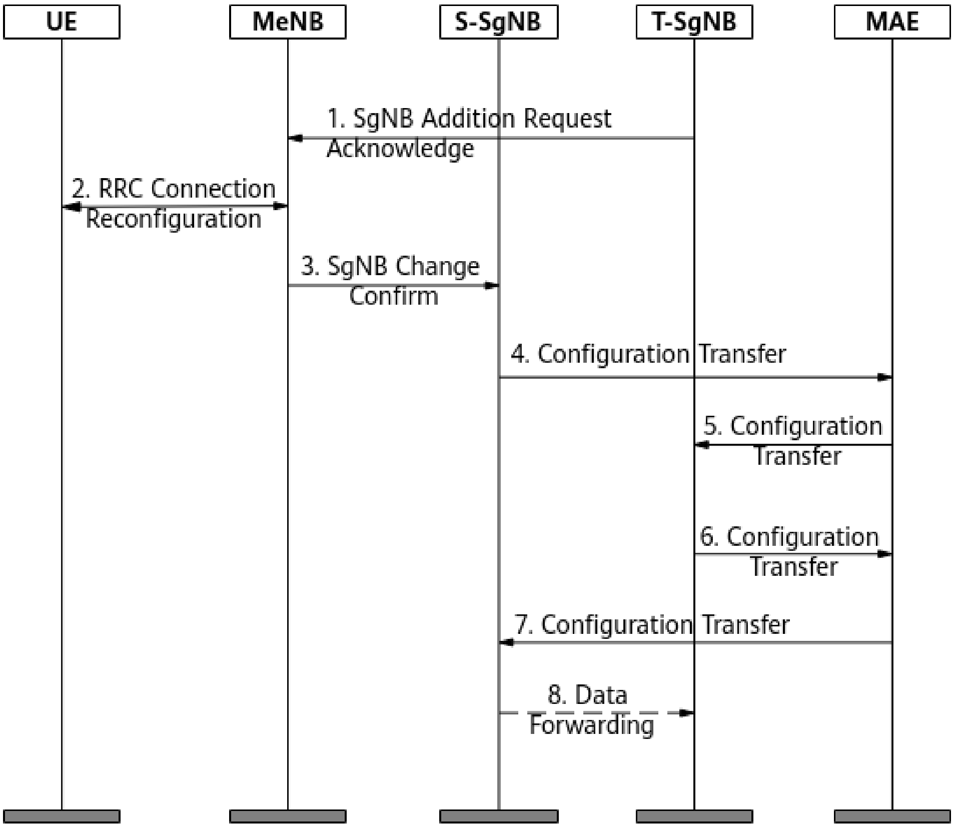
Figure 4-5 Automatic peer configuration between gNodeBs in non-direct IPsec scenarios



The following describes the process of automatic peer configuration and data forwarding between gNodeBs. Other procedures are the same as those for SgNB change. For details, see *NSA Mobility Management*.

1. The target SgNB sends an SgNB Addition Request Acknowledge message to the MeNB, carrying its X2-U IP address.
2. The UE and MeNB exchange RRC Connection Reconfiguration messages.
3. The MeNB forwards the user-plane IP address of the target SgNB to the source SgNB through an SgNB Change Confirm message. The source SgNB automatically generates a **USERPLANEPEER** MO based on the user-plane IP address of the target SgNB in the message and adds the MO to the corresponding **EPGROUP** MO. In this way, a unidirectional X2-U link from the source SgNB to the target SgNB is set up.
4. The source SgNB forwards data to the target SgNB.

Figure 4-6 Automatic peer configuration between gNodeBs in direct IPsec scenarios



The following describes the main process of automatic peer configuration between gNodeBs when direct IPsec is configured for X2-U. For details about direct IPsec setup, see **6.1.1 Direct IPsec Self-Setup**.

1. The target SgNB sends an SgNB Addition Request Acknowledge message to the MeNB, carrying its X2-U IP address.
2. The UE and MeNB exchange RRC Connection Reconfiguration messages.
3. The MeNB forwards the user-plane IP address of the target SgNB to the source SgNB through an SgNB Change Confirm message.
4. If the source SgNB finds no peer information when applying for user-plane resources, it sends a Configuration Transfer message that carries the user-plane IP address and user-plane security IP address to the MAE.
5. The MAE forwards the message to the target SgNB according to the ID of the target SgNB.
6. After receiving the message, the target SgNB sends a Configuration Transfer message that carries its user-plane IP address and user-plane security IP address to the MAE and sets up the X2-U link and user-plane direct IPsec tunnel.
   a. The **USERPLANEPEER** MO is automatically generated based on the user-plane IP address to set up an X2-U link.
   b. The **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model) MOs are automatically generated based on the user-plane security IP address.

(When the **GTRANSPARA**.***TRANSCFGMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **OLD**, the old model is used. When this parameter is set to **NEW**, the new model is used.) In this way, a direct IPsec tunnel is set up for the X2-U interface.

7. The MAE forwards the message to the source SgNB.
8. After receiving the message, the source SgNB sets up an X2-U link and a user-plane direct IPsec tunnel. During an inter-SgNB handover, if an X2-U link and a direct IPsec tunnel have been set up, data is directly forwarded to the target SgNB.
   a. The **USERPLANEPEER** MO is automatically generated based on the user-plane IP address to set up an X2-U link.
   b. The **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model) MOs are automatically generated based on the user-plane security IP address. (When the **GTRANSPARA**.***TRANSCFGMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **OLD**, the old model is used. When this parameter is set to **NEW**, the new model is used.) In this way, a direct IPsec tunnel is set up for the X2-U interface.

For details about MeNB-initiated intra-MeNB handover with an SgNB change and MeNB-initiated inter-MeNB handover with an SgNB change in NSA DC mobility scenarios, see *NSA Mobility Management*. The **GNB_X2U_DATA_FWD_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter can be selected on the eNodeB to support X2-U self-setup only in a direct IPsec scenario for data forwarding during an inter-base-station handover. The MeNB includes the proprietary IEs in the Handover Request Acknowledge and SgNB Release Request messages. The global gNodeB ID of the target gNodeB is carried in the IEs and then sent to the source gNodeB. The source gNodeB obtains the global gNodeB ID of the target gNodeB through the proprietary IE in the SgNB Release Request message. If the source gNodeB determines that direct IPsec is configured, it includes the global gNodeB ID of the target gNodeB in the Configuration Transfer message sent to the MAE in step 4 of automatic peer configuration between gNodeBs in direct IPsec scenarios.

NOTE

- Automatic peer configuration between gNodeBs in direct IPsec scenarios has the following restrictions:
  - Not applicable to base stations managed by different OSSs
  - Not applicable to non-Huawei base stations
- For automatic peer configuration between gNodeBs in direct IPsec scenarios, the source gNodeB starts a 1-hour timer after initiating an X2-U self-setup to a target gNodeB. A second X2-U self-setup to the target gNodeB cannot be initiated before the timer expires to prevent frequent X2-U self-setups between them.
- If both NSA networking and SA networking are configured on a gNodeB, the Xn-U and X2-U interfaces use the same IP address. If the Xn-U interface has been successfully set up, the X2-U and Xn-U interfaces use the same transmission link.

4.1.2.3 Packet Filtering Protected X2 Self-Setup

If integrated firewall is enabled and packet filtering is configured on an eNodeB or a gNodeB, access control list (ACL) rules (defined in the **ACLRULE** MO under the **ACL** MO associated with the **PACKETFILTER** or **PACKETFILTERING** MO) must be configured for automatically established X2-C links and X2-U paths. When the **GTRANSPARA**.***TRANSCFGMODE*** parameter is set to **OLD**, the **PACKETFILTER** MO (old model) is used. When this parameter is set to **NEW**, the **PACKETFILTERING** MO (new model) is used. ACL rules can be manually or automatically configured.

- If the parameter specifying the automatic setup and deletion switch of ACL rules for IP packet filtering (IPv4: **EPGROUP**.***PACKETFILTERSWITCH***; IPv6: **EPGROUP**.***PACKETFILTERSWITCH6***) is set to **ENABLE** for an endpoint group (**EPGROUP** MO) referenced by an X2 object (eNodeB: **X2** MO; gNodeB: **gNBCUX2** MO), ACL rule self-configuration for packet filtering is enabled during X2 self-setup. That is, ACL rules (IPv4: **ACLRULE** MO; IPv6: **ACLRULE6** MO) for the X2-C and X2-U interfaces are automatically created based on the local and peer configurations of the X2 interface. The ACL rules are then automatically added to the corresponding ACL (IPv4: **ACL** MO; IPv6: **ACL6** MO) for packet filtering.
  - If the IDs of ACL rules for packet filtering are manually changed, ACL rules need to be configured manually because the eNodeB or gNodeB does not automatically add the changed ACL rules to the new ACL. If the changed ACL rules are not added to the new ACL, X2-C and X2-U links will be disconnected.
  - If the eNodeB or gNodeB determines that the number of ACL rules has exceeded the maximum number, MML command output will indicate that the number exceeds board capability. In this case, it is recommended that ACL rules be manually configured.
  - For X2 self-setup with peers automatically configured, the target base station cannot obtain the X2-U IP address from the source base station through the data forwarding procedure. Therefore, the ACL rules for packet filtering cannot be automatically configured and the ACL rules for the X2-U interface need to be manually configured.
- If the parameter specifying ACL rule automatic setup and deletion switch for IP packet filtering (IPv4: **EPGROUP**.***PACKETFILTERSWITCH***; IPv6: **EPGROUP**.***PACKETFILTERSWITCH6***) is set to **DISABLE** for an endpoint group (**EPGROUP** MO) referenced by an X2 object (eNodeB: **X2** MO; gNodeB: **gNBCUX2** MO), ACL rules must be manually configured for the peer base station on the eNodeB or gNodeB.

4.1.3 IPv4/IPv6 Dual-Stack Transmission

If an eNodeB or gNodeB supports both IPv4 and IPv6 transmission, the eNodeB or gNodeB supports the dual-stack function. In single-operator scenarios, the X2 user-plane link and X2 control-plane link must use the same IP version. In RAN sharing scenarios, the X2 user-plane link and X2 control-plane link can use different IP versions if they belong to different operators. Either of the following methods can be used to establish an X2 link between the eNodeB and the gNodeB, depending on their configured IP versions:

- X2 self-setup with peers manually configured

  The local and peer ends for an X2 link must use the same IP version (either IPv4 or IPv6). For example, if the local and peer ends use IPv4, the X2 link uses IPv4 as the protocol stack. For details about how to manually configure peers, see **4.1.2.1 X2 Self-Setup with Peers Manually Configured**.

NOTE

In this manual configuration mode, both IPv4 and IPv6 are configured. In this way, there is an attempt to establish two X2 links between local and peer ends. The second X2 link fails to be set up, but double transmission link resources are occupied. Therefore, in this mode, you are advised not to configure both IPv4 and IPv6 at the same time.

- X2 self-setup with peers automatically configured

  Based on the local IP stack (IPv4/IPv6) and the IP version preference (specified by **EPGROUP**.***IPVERPREFERENCE (5G gNodeB, LTE eNodeB)***) configured on the eNodeB and gNodeB, X2 links with different IP versions are automatically established between them, as described in **Table 4-2**. For details about how to automatically configure peers, see **4.1.2.2 X2 Self-Setup with Peers Automatically Configured**.

  If the IPv4/IPv6 dual-stack transmission is used over the X2 interfaces between base stations, the X2 interfaces must use the same IP version, which is specified by the **EPGROUP**.***IPVERPREFERENCE (5G gNodeB, LTE eNodeB)*** parameter.

Table 4-2 Rules of configuring the IP version for the X2 link between the eNodeB and the gNodeB

IP Version for the eNodeB	IP Version for the gNodeB	IP Version Preference for the gNodeB	IP Version for the X2 Link
IPv4	IPv4	N/A	IPv4
IPv6	IPv6	N/A	IPv6
IPv4/IPv6	IPv4	N/A	IPv4
IPv4/IPv6	IPv6	N/A	IPv6
IPv4	IPv4/IPv6	N/A	IPv4
IPv6	IPv4/IPv6	N/A	IPv6
IPv4/IPv6	IPv4/IPv6	IPv4	IPv4
IPv4/IPv6	IPv4/IPv6	IPv6	IPv6

NOTE

- A change in the **EPGROUP**.***IPVERPREFERENCE (5G gNodeB, LTE eNodeB)*** parameter setting will not cause the control-plane or user-plane X2 link to be reestablished. The parameter setting will be checked only when X2 setup or reestablishment is required.
- If both IPv4 and IPv6 addresses are configured on the eNodeB for which automatic peer configuration is used, the eNodeB sends two IP addresses to the peer gNodeB. If the peer gNodeB is configured with only one IP address, the IP version of this IP address is used for X2 link setup. If the peer gNodeB is configured with both IPv4 and IPv6 addresses, the IP version used for X2 link setup is determined by the **EPGROUP**.***IPVERPREFERENCE (5G gNodeB, LTE eNodeB)*** parameter setting on the gNodeB.

4.2 X2 Self-Update

X2 self-update of peer information can be service-triggered, or it can be triggered manually.

4.2.1 Service-Triggered Self-Update of X2 Peer Information

During network operation, X2 self-setup will be triggered and **SCTPPEER** and **USERPLANEPEER** MOs of the X2 interface will be automatically updated when the following conditions apply: (1) The IP address or operator information of the peer is changed, such as the mobile network code (MNC), mobile country code (MCC), or global eNodeB/gNodeB ID; (2) the X2 interface is faulty; (3) UEs initiate DC services.

Change in the IP Address of the Peer

If the control-plane IP address of the peer is changed, the X2 interface with the peer will become faulty. If a DC service is initiated when the X2 interface is faulty, X2 self-setup will be triggered. The local end obtains the new control-plane IP address of the peer during the self-setup. The local end searches for an **SCTPPEER** MO that contains the same operator information (indicated by **SCTPPEER**.***REMOTEID (LTE eNodeB, 5G gNodeB)***) as the operator information (such as MNC, MCC, and global eNodeB/gNodeB ID) in the Configuration Transfer message. The local end then updates the control-plane IP address in the **SCTPPEER** MO.

If the user-plane IP address of the peer is changed, the X2 interface with the peer will become faulty. If a DC service is initiated when the X2 interface is faulty, X2 self-setup will be triggered. The local end obtains the new user-plane IP address of the peer during the self-setup. The local end searches for a **USERPLANEPEER** MO that contains the same operator information (indicated by **USERPLANEPEER**.***REMOTEID (LTE eNodeB, 5G gNodeB)***) as the operator information (MNC, MCC, and global eNodeB/gNodeB ID) in the Configuration Transfer message. The local end then generates a new **USERPLANEPEER** MO based on the modified user-plane IP address of the peer. The faulty user-plane link leads to the reporting of ALM-25954 User Plane Fault and is deleted through X2 self-removal (see **4.3 X2 Self-Removal** for details).

NOTE

Based on information about the local and peer endpoints included in the **EPGROUP** MO, the eNodeB or gNodeB can automatically generate the **SCTPLNK** and **CPBEARER** MOs with the **SCTPLNK**.***CTRLMODE (LTE eNodeB, 5G gNodeB)*** and **CPBEARER**.***CTRLMODE (LTE eNodeB, 5G gNodeB)*** parameters set to **AUTO_MODE**. Do not manually modify or remove these automatically generated MOs. Otherwise, the corresponding X2 interface cannot work properly. If an X2 interface fails to work properly due to the preceding reason, remove the **X2** MO and then add it again.

Change in the Operator Information of the Peer

If the operator information (such as MNC, MCC, and global gNodeB ID) of the peer is changed, the X2 interface does not become faulty, and the values of the **SCTPPEER**.***REMOTEID (LTE eNodeB, 5G gNodeB)*** and

USERPLANEPEER.***REMOTEID (LTE eNodeB, 5G gNodeB)*** parameters do not automatically update since X2 self-setup is not triggered.

X2 self-setup is triggered only when a UE initiates DC services in the case that the X2 interface is faulty due to a transmission link fault or any other reason. During this process, the local end searches for peers (specified by the **SCTPPEER** and **USERPLANEPEER** MOs) that match the peer IP address information contained in the Configuration Transfer message. The local end then updates the values of the corresponding **SCTPPEER**.***REMOTEID (LTE eNodeB, 5G gNodeB)*** and **USERPLANEPEER**.***REMOTEID (LTE eNodeB, 5G gNodeB)*** parameters.

Change in the IP Address and Operator Information of the Peer

If both the IP address and operator information of the peer on the X2 interface are changed, the local end cannot obtain the peer information, and therefore does not automatically update the **SCTPPEER** or **USERPLANEPEER** MO.

Change in the SCTP Multi-Homing IP Addresses of the Peer

Peer SCTP multi-homing reconstruction involves the following scenarios:

- Scenario 1: Both IP addresses of the control-plane endpoint of the peer are changed.
- Scenario 2: A second IP address is added to the control-plane endpoint of the peer, so that the endpoint becomes a multi-homed endpoint.
- Scenario 3: The control-plane endpoint of the peer is multi-homed, and only one of its IP addresses is changed.

In scenario 1, the X2 interface between the local and peer ends becomes faulty. In this case, X2 self-setup is triggered only when a UE initiates a handover to the target gNodeB. If this happens, configurations of the **SCTPPEER** MO are updated. For details, see **Change in the IP Address of the Peer**.

In scenarios 2 and 3, the SCTP link to the local base station still works properly as one IP address of the peer SCTP multi-homed endpoint remains unchanged. The peer base station sends an SCTP INIT message containing the updated peer IP address list to the local base station. When the X2 interface works properly and the local base station determines that the IP addresses in the IP address list carried in the SCTP INIT message differ from those in the **SCTPPEER** MO, the local and peer base stations exchange Huawei proprietary messages (huawei_Private_MSG) containing their respective configuration information to update the configuration of the **SCTPPEER** MO.

4.2.2 Manually-Triggered Self-Update of X2 Peer Information

For the X2 interface, the IP addresses of the local base station are referenced in the **SCTPPEER** and **USERPLANEPEER** MOs of the peer base station. When the IP addresses of the local base station change, the MAE can automatically associate and change the **SCTPPEER** and **USERPLANEPEER** MOs of the peer base station that has the same routing domain (**VRF** MO) as the local base station.

4.3 X2 Self-Removal

X2 Self-Removal Triggered by Link Faults

- If the **EnodebAlgoExtSwitch**.***X2SonDelTimerForLNX2Fault*** parameter is set to a value other than **0** on the LTE side, a timer specified by this parameter starts when an X2 control-plane link fault occurs. If the fault persists until the timer expires, local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed. The timer starts 10 minutes after the local base station starts up. This prevents the X2 interface from being mistakenly removed when the timer expires during the local base station startup.

  It is recommended that the **EnodebAlgoExtSwitch**.***X2SonDelTimerForLNX2Fault*** parameter be set to a value specifying a period for at least 10 minutes. If this period is extremely short, the X2 interface can be mistakenly removed.
- If an X2 control-plane link fault occurs after the **X2SON_DEL_FOR_X2FAULT_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter is selected on the NR side, an exception timer starts. If the fault persists throughout the period specified by the **gNBX2SonConfig**.***X2SonDeleteTimerForX2Fault*** parameter on the NR side, local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed.

NOTE

In the case of hybrid SA and NSA networking where the user planes of the X2 and Xn interfaces or those of the X2 and eXn interfaces use the same IP address, to ensure that services on the Xn or eXn interface are not affected, the peer of the X2-U interface is not removed when X2 self-removal is triggered.

X2 Self-Removal Triggered by Immediate Faults After Initial X2 Self-Setup

An X2 interface is automatically removed when all the following conditions are met:

- The X2 interface is initially set up, and the SCTP link is faulty.
- On the LTE and NR sides, the corresponding switches are turned on:
  - The **EnodebAlgoExtSwitch**.***GnbX2InitFailDelSwitch*** parameter is set to **ON** on the LTE side. The eNodeB identifies a link that has a fault immediately after initial X2 self-setup is completed. After the initial fault lasts for several minutes (less than 10 minutes), the eNodeB automatically removes and blacklists this X2 link. For details, see **4.4.2 Dynamic X2 Blacklist**.
  - The **X2INIT_FAIL_DEL_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter is selected on the NR side. The gNodeB identifies a link that has a fault immediately after initial X2 self-setup is completed. After the fault lasts for several minutes (less than 10 minutes), the gNodeB automatically removes this link.
- The **SCTPPEER** MO is automatically configured.

When the preceding conditions are met:

- If the **SCTPPEER**.***CTRLMODE*** parameter is set to **AUTO_MODE**, the X2 interface and the **SCTPLNK** and **SCTPPEER** MOs are removed.
- If the **SCTPPEER**.***CTRLMODE*** parameter is set to **MANUAL_MODE**, the X2 interface and the **SCTPLNK** MO are removed. The **SCTPPEER** MO is not removed.

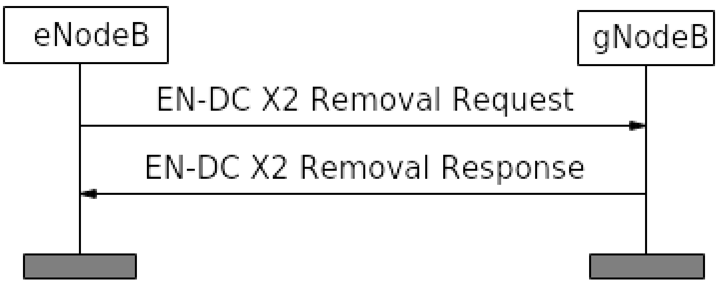
NOTE

In the case of hybrid SA and NSA networking where the user planes of the X2 and Xn interfaces use the same IP address, to ensure that services on the Xn interface are not affected, the peer of the X2-U interface is not removed when X2 self-removal is triggered.

X2 Self-Removal Based on X2 Usage

- On the LTE side, if the value of the **EnodebAlgoExtSwitch**.***X2SonDelTimerForLNX2Usage*** parameter is not 0, the number of successful SgNB additions over an X2 interface is less than or equal to the value of the **EnodebAlgoExtSwitch**.***X2SonDeleteGnbAddCntThld*** parameter within the measurement period specified by the **EnodebAlgoExtSwitch**.***X2SonDelTimerForLNX2Usage*** parameter, and there are no UEs performing NSA services, then X2 self-removal based on X2 usage is triggered. Local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed. When the **SCTPPEER**.***CTRLMODE*** and **USERPLANEPEER**.***CTRLMODE*** parameters are set to **AUTO_MODE**, automatically created MOs (such as the **X2INTERFACE** MO) are also removed.

  Figure 4-7 LTE-triggered X2 self-removal based on X2 usage

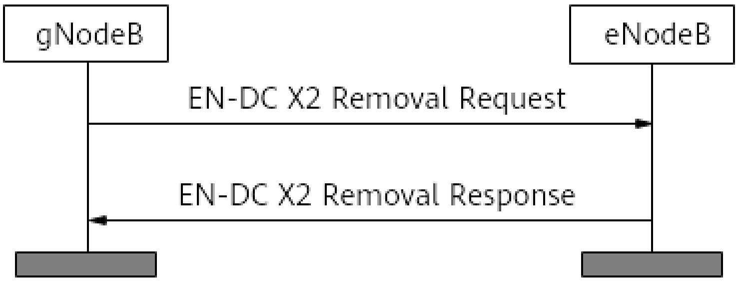
  

  In **Figure 4-7**:

  a. The eNodeB sends an EN-DC X2 Removal Request message to the peer gNodeB, instructing the gNodeB to remove its X2 interface. The gNodeB sends an EN-DC X2 Removal Response message and removes its X2 interface.
  b. After receiving the EN-DC X2 Removal Response message, the eNodeB removes its X2 interface.

- On the NR side, if the **X2SON_DEL_FOR_X2USAGE_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter is selected, the number of successful SgNB additions over an X2 interface is less than or equal to the value of the **gNBX2SonConfig**.***X2SonDeleteGnbAddCntThld*** parameter within the measurement period specified by the **gNBX2SonConfig**.***X2SonDeleteTimerForX2Usage*** parameter, and there are no UEs performing NSA services, then X2 self-removal based on X2 usage is triggered. Local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed. When the **SCTPPEER**.***CTRLMODE*** and **USERPLANEPEER**.***CTRLMODE*** parameters are set to **AUTO_MODE**, automatically created MOs (such as the **X2INTERFACE** MO) are also removed.

Figure 4-8 NR-triggered X2 self-removal based on X2 usage



In **Figure 4-8**:

a. The gNodeB sends an EN-DC X2 Removal Request message to the peer eNodeB, instructing the eNodeB to remove its X2 interface. The eNodeB sends an EN-DC X2 Removal Response message and removes its X2 interface.
b. After receiving the EN-DC X2 Removal Response message, the gNodeB removes its X2 interface.

NOTE

- Before enabling the X2 self-removal function on the LTE or NR side, ensure that the X2 self-setup switch is turned on. Otherwise, NSA services may fail.
- The remote ID of the X2 interface to be removed must comply with the following format: "eNB:MCC_xxx MNC_xxx GlobaleNBId_xxx-xxx-xxxxxxx" (for an eNodeB) or "gNB:MCC_xxx MNC_xxx GlobalgNBId_xxx-xxx-xxxxxxx" (for a gNodeB).
- A smaller value of the **gNBX2SonConfig**.***X2SonDeleteTimerForX2Usage*** parameter results in a higher probability of meeting the conditions for X2 link removal. As a result, X2 links are quickly removed and then automatically set up, causing frequent X2 link additions and removals. The period must be longer than at least one day when fast and automatic adjustment of the X2 interface is required.
- In the case of hybrid SA and NSA networking where the user planes of the X2 and Xn interfaces use the same IP address, to ensure that services on the Xn interface are not affected, the peer of the X2-U interface is not removed when X2 self-removal is triggered.
- When X2 self-removal based on X2 usage is performed, an X2 link is not removed if this link is required by inter-base-station coordination services.

X2-U Peer Self-Removal

When the **USERPLANEPEER**.***CTRLMODE*** parameter is set to **AUTO_MODE** on the eNodeB or gNodeB side, the eNodeB or gNodeB can automatically remove the X2-U transmission links and the **USERPLANEPEER** MOs. When this parameter is set to **MANUAL_MODE**, the eNodeB or gNodeB does not automatically remove the X2-U transmission links and the **USERPLANEPEER** MOs.

When the **GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)*** parameter is set to **ENABLE**, the eNodeB or gNodeB determines whether to automatically remove the X2-U transmission links in either of the following situations:

- Self-removal based on link faults

  If the GTP-U detection result indicates that all the transmission links corresponding to **USERPLANEPEER** MOs are faulty, the eNodeB or gNodeB periodically (every 60 minutes) attempts to remove those transmission links (up to 50 links at one time) that remain faulty for over one hour and the corresponding **USERPLANEPEER** MOs.
- Self-removal based on the aging mechanism
  - In case of an excess over the specifications of a base station

    The eNodeB or gNodeB marks a transmission link as aged and processes as follows if the link is not used by any service.

    - The base station performs self-removal based on the aging mechanism once every hour and checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the base station removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs. If it does not, no aged transmission links are removed.
    - When new transmission links need to be established, the base station checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the base station removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs, and establishes new transmission links. If it does not, no aged transmission links are removed.
    - If the maximum allowed number of transmission links is reached and there is no aged link, the eNodeB or gNodeB rejects new UEs' access requests.
  - In case of an excess over the specifications of a single endpoint group

    "Self-removal in case of an excess over the specifications of a single endpoint group" takes effect for the eNodeB or gNodeB when the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (5G gNodeB, LTE eNodeB)*** parameter is selected. Self-removal based on the aging mechanism starts when the number of peers in a single endpoint group exceeds 90% of the maximum number allowed by the in-use board. The condition for marking a link as aged and the self-removal mechanism are the same as those for self-removal in case of an excess over the specifications of a base station.

NOTE

In the case of hybrid SA and NSA networking where the user planes of the X2 and Xn interfaces use the same IP address, to ensure that services on the Xn interface are not affected, the peer of the X2-U interface is not removed when X2 self-removal is triggered.

4.4 X2 Blacklist and X2 Whitelist

4.4.1 Static X2 Blacklist and Whitelist

Operators can run the **ADD ENDCX2BLACKWHITELIST** command on the eNodeB to add a peer gNodeB to an EN-DC X2 blacklist or whitelist.

If a peer gNodeB is on the EN-DC X2 blacklist, an X2 link between the eNodeB and this gNodeB cannot be automatically set up. If the peer gNodeB that has set up an X2 link with the eNodeB is added to the EN-DC X2 blacklist, the X2 link will become faulty and cannot be automatically removed. It is recommended that this X2 link be manually removed.

If a peer gNodeB is on the EN-DC X2 whitelist, an X2 link between the eNodeB and this gNodeB cannot be automatically removed.

NOTE

If the ID of a blacklisted or whitelisted gNodeB is changed, you need to manually update the value of the **EnDcX2BlackWhiteList**.***gNodeBId*** parameter. Otherwise, if the X2 link set up with this gNodeB is not removed, the blacklist or whitelist function still takes effect for the gNodeB.

A maximum of 128 EN-DC X2 blacklist and whitelist entries can be set.

4.4.2 Dynamic X2 Blacklist

An X2 interface cannot be automatically set up when the local and peer base stations of the X2 interface are in the dynamic blacklist and the gNodeB X2 dynamic blacklist aging timer (specified by the **EnodebAlgoExtSwitch**.***GnbX2DynBlklistAgingTimer*** parameter) has not expired. Specifically, the local eNodeB neither initiates a self-setup request for the X2 interface nor responds to the X2 self-setup request initiated by the peer gNodeB. When the timer expires, the local eNodeB removes the peer gNodeB from the corresponding blacklist.

The dynamic X2 blacklist specification is the same as the X2 interface specification. The specification is shared by LTE-LTE X2 and LTE-NR X2 interfaces. In the multi-MPT co-BBU scenario, the dynamic X2 blacklist specification is the same as the maximum X2 interface specification among main control boards. If the number of entries in a dynamic blacklist has reached the maximum and a new entry needs to be added, the new entry will replace the one with the shortest remaining aging time.

Operators can run the **CLR X2DYNBLACKLIST** command on the LTE side to remove a peer gNodeB from the dynamic X2 blacklist before the gNodeB X2 dynamic blacklist aging timer expires.

4.5 Application Restrictions

For an operator, it is recommended that only one **gNBCUX2** MO be configured on the gNodeB. If multiple **gNBCUX2** MOs are configured for an operator, only the one with the smallest **gNBCUX2**.***gNBCuX2Id*** value takes effect.

4.6 Network Analysis

4.6.1 Benefits

X2 self-management in EN-DC scenarios simplifies configuration operations, reduces operating expense (OPEX) for operators, and improves eNodeB/gNodeB resource usage.

4.6.2 Impacts

The impacts between this function and other functions are described in **4.7.2.3 Function Impacts**.

When the mis-deletion prevention mechanism for X2 self-removal triggered by link faults is in effect, the number of X2 interface setup attempts at the local eNodeB (indicated by the **L.Sig.X2.SendSetup.Att** counter) and the number of successful X2 interface setups at the local eNodeB (indicated by the **L.Sig.X2.SendSetup.Succ** counter) may decrease.

When the number of X2 interfaces is insufficient, if the number of remaining X2 interfaces is limited due to X2 self-setup triggered by inter-base-station coordination services, the values of the counters related to handovers and number of UEs may be affected. The following lists typical counters.

Counter Name	Possible Impact
N.NsaDc.InterSgNB.PSCell.Change.Succ	Decrease
N.NsaDc.InterSgNB.SgNB.Add.Succ	Decrease
N.NsaDc.InterSgNB.SgNB.Add.Ack	Decrease
N.NsaDc.InterSgNB.PSCell.Change.Fail.Trans	Increase
N.NsaDc.SgNB.Add.Succ	Decrease
N.User.NsaDc.PSCell.Avg	Decrease
N.User.NsaDc.PSCell.Max	
L.HHO.X2.IntraFreq.PrepAttOut	Decrease
L.HHO.X2.InterFreq.PrepAttOut	
L.HHO.X2.InterFddTdd.PrepAttOut	
L.HHO.X2.IntraFreq.Succ.ReEst2Src	
L.HHO.X2.InterFreq.Succ.ReEst2Src	
L.HHO.X2.PrepAttIn	
L.HHO.S1.NoCntxReEst2Other.Success	Increase
L.Traffic.User.NsaDc.PCell.Avg	Decrease
L.Traffic.User.NsaDc.PCell.Max	

Counter Name	Possible Impact
L.NsaDc.SgNB.Add.Att	Decrease
L.NsaDc.SgNB.Add.Succ	
L.NsaDc.SCG.Change.Att	
L.NsaDc.SgNB.Add.Req.Ack	
L.NsaDc.SgNB.Add.Succ	
L.NsaDc.SCG.Change.Succ	

4.7 Requirements

4.7.1 Licenses

None

4.7.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.7.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Automatic user plane removal	**GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)***	*X2 and S1 Self-Management in NSA Networking*	Ensure that the **GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)*** parameter is set to **ENABLE** before enabling "self-removal in case of an excess over the specifications of a single endpoint group" (by selecting the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)*** parameter).
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Self-removal in case of an excess over the specifications of a single endpoint group	**SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)*** parameter	*X2 and S1 Self-Management in NSA Networking*	Ensure that "self-removal in case of an excess over the specifications of a single endpoint group" is disabled before disabling automatic user plane removal (by setting the **GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)*** parameter to **DISABLE**).

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RAN sharing with common carrier	**ENodeBSharingMode**.***ENodeBSharingMode*** set to **SHARED_FREQ**	*RAN Sharing* in *eRAN Feature Documentation*	The **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NetworkPrfmOptSwitch*** parameter can be selected only when the **ENodeBSharingMode**.***ENodeBSharingMode*** parameter is set to **SHARED_FREQ**.

4.7.2.2 Mutually Exclusive Functions

None

4.7.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	EN-DC X2 selection optimization	**HYBRID_SHR_ENDC_X2_SEL_OPT_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter	*RAN Sharing* in *eRAN Feature Documentation*	In NSA networking, when EN-DC X2 selection optimization is enabled, an operator cannot use the X2 interface of another carrier-sharing operator group. If there is no matched LTE-NR X2 interface and the X2 self-setup function is disabled, no X2 interface is available.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	X2 interface self-setup based on target operators	**GlobalProcSwitch**.***TargetOpBasedX2Switch***	None	When the switch controlling X2 interface self-setup based on target operators is turned on, the eNodeB can automatically set up X2 interfaces only with eNodeBs or gNodeBs of a specified operator.

4.7.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For LTE, a UMPT is configured as the main control board.

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

N/A

4.7.4 Networking

LTE and NR Intra-Base-Station X2 Transmission Networking

In NSA networking, X2 interface data between LTE and NR deployed on the same base station can be transmitted through the backplane and CI interconnection cables to minimize transmission latency over the X2 interface between LTE and NR. This applies when LTE and NR are deployed in the same BBU, LTE and NR are deployed in different BBUs, or LTE and NR share the same main control board.

LTE and NR co-site co-BBU or separate-BBU scenarios have the following requirements on the X2 transmission configuration:

- The X2 interface IP addresses, backplane tunnels, and routes to the peer IP addresses are configured on both the LTE and NR sides.
- Intra-base-station X2 user-plane data can be forwarded through internal addresses. To achieve this, the **gNodeBParam**.***X2uTransmissionType*** parameter must be set to **INTRA_TRANS**, but IP routes are not required for the X2 user plane. In a separate-MPT base station, for X2 self-setup with peers manually configured, the peer gNodeB/eNodeB does not need to be configured as the user-plane peer (**USERPLANEPEER** MO) of the local eNodeB/gNodeB.
- The X2 interface does not use source-based routing. If the S1 interface uses source-based routing and its interface data is transmitted through panel interconnection, the S1 interface must use a different IP address from the X2 interface. This is because source-based routes take precedence over destination-based routes. If a source-based route is configured, all packets that match the source-based route are forwarded using the source-based route, not the destination-based route.

NOTE

Intra-base-station X2 transmission does not support IPsec.

In co-MPT scenarios, requirements on the X2 transmission configuration are as follows:

- X2 interface IP addresses must be configured on both the LTE and NR sides. IP routes are not required for intra-base-station X2 transmission.
- Intra-base-station X2 user-plane data can only be forwarded through internal addresses. To achieve this, the **gNodeBParam**.***X2uTransmissionType*** parameter must be set to **INTRA_TRANS**, but IP routes are not required for the X2 user plane. In a co-MPT base station, for X2 self-setup with peers manually configured, the peer gNodeB/eNodeB does not need to be configured as the user-plane peer (**USERPLANEPEER** MO) of the local eNodeB/gNodeB; for X2 self-setup with peers automatically configured, the user-plane peer is not generated automatically.

In a separate-MPT base station with inter-BBU CI interconnection, the user-plane data of X2 interfaces between LTE and NR can be transmitted through the IP network. The **gNodeBParam**.***X2uTransmissionType*** parameter must be set to **ROUTING_TRANS**. Other configurations are the same as those in **LTE and NR Inter-Base-Station X2 Transmission Networking and NR and NR Inter-Base-Station X2-U Transmission Networking**.

NOTE

- The new setting takes effect 18s after the **gNodeBParam**.***X2uTransmissionType*** parameter is modified.
- Modification to the **gNodeBParam**.***X2uTransmissionType*** parameter applies only to UEs that newly access the network.

LTE and NR Inter-Base-Station X2 Transmission Networking and NR and NR Inter-Base-Station X2-U Transmission Networking

X2 interface data can be transmitted through the IP networks when the eNodeB and gNodeB are not co-sited or when X2-U transmission exists between gNodeBs. The IP address of the X2 user-plane interface must be configured for transmitting user-plane data between the base stations.

NSA and SA Hybrid Networking

If the X2 and Xn interfaces need to share an IP address, the X2-C and Xn-C interfaces must have different SCTP port numbers, and the X2-U and Xn-U interfaces must share the same **EPGROUP**.

4.7.5 Others

The base stations on both sides of the X2 interface must be Huawei base stations.

CI interconnection and IP transmission interconnection are supported between the eNodeB and gNodeB. When both CI interconnection and IP transmission interconnection are configured, CI interconnection is used preferentially for user-plane transmission to reduce latency, and IP transmission interconnection is used for control-plane transmission. When only CI interconnection is configured between the eNodeB and gNodeB, CI interconnection is used on both the control and user planes for transmission.

4.8 Operation and Maintenance

4.8.1 Data Configuration

4.8.1.1 Data Preparation

Configuring Common Parameters

Common parameters must be configured at both the local and peer ends.

The following table describes the parameters that must be set in an **SCTPTEMPLATE** MO.

Parameter Name	Parameter ID	Setting Notes
SCTP Parameters Template ID	**SCTPTEMPLATE**.***SCTPTEMPLATEID* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
DSCP Switch	**SCTPTEMPLATE**.***DSCPSW* (5G gNodeB, LTE eNodeB)**	The default value is recommended.

The following table describes the parameters that must be set in an **SCTPHOST** MO.

Parameter Name	Parameter ID	Setting Notes
SCTP Host ID	**SCTPHOST**.***SCTPHOSTID* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
IP Version	**SCTPHOST**.***IPVERSION* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
First Local IP Address	**SCTPHOST**.***SIGIP1V4* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
First Local IPv6 Address	**SCTPHOST**.***SIGIP1V6* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
Second Local IP Address	**SCTPHOST**.***SIGIP2V4* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
Second Local IPv6 Address	**SCTPHOST**.***SIGIP2V6* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
Local SCTP Port No.	**SCTPHOST**.***PN* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.
SCTP Parameters Template ID	**SCTPHOST**.***SCTPTEMPLATEID* (5G gNodeB, LTE eNodeB)**	Set this parameter based on the network plan.

(Optional, required for manual peer configuration) The following table describes the parameters that must be set in an **SCTPPEER** MO.

Parameter Name	Parameter ID	Setting Notes
IP Version	**SCTPPEER**.***IPVERSION (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
First Peer IP Address	**SCTPPEER**.***SIGIP1V4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
First Peer IPv6 Address	**SCTPPEER**.***SIGIP1V6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Second Peer IP Address	**SCTPPEER**.***SIGIP2V4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Second Peer IPv6 Address	**SCTPPEER**.***SIGIP2V6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Peer SCTP Port No.	**SCTPPEER**.***PN (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	**USERPLANEHOST**.***UPHOSTID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
IP Version	**USERPLANEHOST**.***IPVERSION (5G gNodeB, LTE eNodeB)***	● If the peer uses IPv4, set this parameter to **IPv4**. ● If the peer uses IPv6, set this parameter to **IPv6**.
Local IP Address	**USERPLANEHOST**.***LOCIPV4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Local IPv6 Address	**USERPLANEHOST**.***LOCIPV6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

(Optional, required for manual peer configuration) The following table describes the parameters that must be set in a **USERPLANEPEER** MO.

Parameter Name	Parameter ID	Setting Notes
User Plane Peer ID	**USERPLANEPEER**.***UPPEERID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
IP Version	**USERPLANEPEER**.***IPVERSION (5G gNodeB, LTE eNodeB)***	• If the peer uses IPv4, set this parameter to **IPv4**. • If the peer uses IPv6, set this parameter to **IPv6**.
Peer IP Address	**USERPLANEPEER**.***PEERIPV4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Peer IPv6 Address	**USERPLANEPEER**.***PEERIPV6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **EPGROUP** MO. It is recommended that the **USERPLANEHOST** and **USERPLANEPEER** MOs or the **SCTPHOST** and **SCTPPEER** MOs for an operator be configured in the same **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	**EPGROUP**.***EPGROUPID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
SCTP Host List	**EPGROUP**.***SCTPHOSTLIST (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
SCTP Peer List	**EPGROUP**.***SCTPPEERLIST (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan. If there are multiple peers, multiple SCTP peers exist in the list specified by the **EPGROUP**.***SCTPPEERLIST (5G gNodeB, LTE eNodeB)*** parameter.
User Plane Host List	**EPGROUP**.***USERPLANEHOSTLIST (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
User Plane Peer List	**EPGROUP**.***USERPLANEPEERLIST (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
IP Protocol Version Preference	**EPGROUP**.***IPVERPREFERENCE (5G gNodeB, LTE eNodeB)***	This parameter specifies the preferred IP version for automatic link setup when both IPv4 and IPv6 addresses are configured on the local and peer ends.
Packet Filter ACL Rule Auto-Setup-Deletion SW	**EPGROUP**.***PACKETFILTERSWITCH (5G gNodeB, LTE eNodeB)***	This parameter specifies whether to enable the ACL rule self-configuration function for packet filtering.
IPv6 Packet Filter ACL Rule Auto-Setup-Deletion SW	**EPGROUP**.***PACKETFILTERSWITCH6 (5G gNodeB, LTE eNodeB)***	This parameter specifies whether to enable the ACL rule self-configuration function for IPv6 packet filtering.

Configuring Parameters on the LTE Side

The following table describes the parameters that must be set in the **GlobalProcSwitch** and **EnodebAlgoExtSwitch** MOs.

Parameter Name	Parameter ID	Option	Setting Notes
Interface Setup Policy Switch	**GlobalProcSwitch**.***InterfaceSetupPolicySw***	**LTE_NR_X2_SON_SETUP_SW**	Set this parameter based on the network plan.
X2 Setup Failure Num Threshold	**GlobalProcSwitch**.***X2SetupFailureNumThd***	N/A	Set this parameter based on the network plan.
X2 Setup Failure Num Timer	**GlobalProcSwitch**.***X2SetupFailureNumTimer***	N/A	Set this parameter based on the network plan.
gNodeB X2 Initial Fail Delete Switch	**EnodebAlgoExtSwitch**.***GnbX2InitFailDelSwitch***	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
X2 SON Delete Timer for LTE-NR X2 Fault	**EnodebAlgoExtSwitch**.***X2SonDelTimerForLNX2Fault***	N/A	Set this parameter based on the network plan. It is recommended that this parameter be set to a value greater than or equal to **10** if X2 self-removal triggered by link faults is required.
X2 SON Delete Timer for LTE-NR X2 Usage	**EnodebAlgoExtSwitch**.***X2SonDelTimerForLNX2Usage***	N/A	Set this parameter based on the network plan.
X2 SON Delete gNodeB Addition Count Threshold	**EnodebAlgoExtSwitch**.***X2SonDeleteGnbAddCntThld***	N/A	Set this parameter based on the network plan.
gNodeB X2 Dynamic Blacklist Aging Timer	**EnodebAlgoExtSwitch**.***GnbX2DynBlklistAging-Timer***	N/A	Set this parameter based on the network plan.

The following table describes the parameters that must be set to configure the reference relationship between the **X2** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
X2 ID	**X2**.***X2Id***	Set this parameter based on the network plan.
CN Operator ID	**X2**.***CnOperatorId***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
End Point Group Config Flag	**X2**.***EpGroupCfgFlag***	This parameter specifies whether to configure the control-plane or user-plane endpoint group. ● Set this parameter to **CP_CFG** when only the control-plane **EPGROUP** MO is referenced by the **X2** MO. ● Set this parameter to **UP_CFG** when only the user-plane **EPGROUP** MO is referenced by the **X2** MO. ● Set this parameter to **CP_UP_CFG** when both control-plane and user-plane **EPGROUP** MOs are referenced by the **X2** MO.
Control Plane End Point Group ID	**X2**.***CpEpGroupId***	Set this parameter based on the network plan.
User Plane End Point Group ID	**X2**.***UpEpGroupId***	Set this parameter based on the network plan.
Peer Base Station Release	**X2**.***TargetENodeBRelease***	This parameter specifies the protocol release with which the peer eNodeB connected to the local base station through the X2 interface complies.

The following table describes the parameters that must be set in an **EnDcX2BlackWhiteList** MO on the eNodeB side.

Parameter Name	Parameter ID	Setting Notes
Mobile country code	**EnDcX2BlackWhiteList**.***Mcc***	Set this parameter based on the network plan.
Mobile network code	**EnDcX2BlackWhiteList**.***Mnc***	Set this parameter based on the network plan.
gNodeB ID	**EnDcX2BlackWhiteList**.***gNodeBId***	Set this parameter based on the network plan.
List Type	**EnDcX2BlackWhiteList**.***ListType***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **SCTPHOST** MO.

Parameter Name	Parameter ID	Setting Notes
Simple Mode Switch	**SCTPHOST**.***SIMPLEMODESWITCH (LTE eNodeB, 5G gNodeB)***	You are advised to deselect this option in LTE and NR co-MPT scenarios on an NSA network.

The following table describes the parameters that must be set in an **EnodebAlgoExtSwitch** MO.

Parameter Name	Parameter ID	Setting Notes
Simple Mode Switch	**EnodebAlgoExtSwitch**.***NetworkPrfmOptSwitch***	It is recommended that the **SHARE_FREQ_X2_UP_OPT_SW** option be selected if X2-U link self-setup/self-removal procedures need to be separated for different operators in NSA networking.

Configuring Parameters on the NR Side

The following table describes the parameters that must be set in a **gNBX2SonConfig** MO.

Parameter Name	Parameter ID	Option	Setting Notes
X2 SON Configuration Switch	**gNBX2SonConfig**.***X2SonConfigSwitch***	**X2SON_SETUP_SWITCH**	Set this parameter based on the network plan.
X2 SON Configuration Switch	**gNBX2SonConfig**.***X2SonConfigSwitch***	**X2INIT_FAIL_DEL_SWITCH**	Set this parameter based on the network plan.
X2 SON Configuration Switch	**gNBX2SonConfig**.***X2SonConfigSwitch***	**X2SON_DEL_FOR_X2FAULT_SWITCH**	**gNBX2SonConfig**.***X2SonDeleteTimerForX2Fault*** specifies the timer for X2 self-removal based on link faults.

Parameter Name	Parameter ID	Option	Setting Notes
X2 SON Configuration Switch	**gNBX2SonConfig**.***X2SonConfigSwitch***	**X2SON_DEL_FOR_X2USAGE_SWITCH**	**gNBX2SonConfig**.***X2SonDeleteGnbAddCntThld*** specifies the threshold for X2 self-removal based on X2 usage.
X2 SON Delete Timer for X2 Fault	**gNBX2SonConfig**.***X2SonDeleteTimerForX2Fault***	N/A	This is a timer for X2 self-removal based on link faults.
X2 SON Delete gNodeB Addition Count Threshold	**gNBX2SonConfig**.***X2SonDeleteGnbAddCntThld***	N/A	Set this parameter based on the network plan.
X2 SON Delete Timer for X2 Usage	**gNBX2SonConfig**.***X2SonDeleteTimerForX2Usage***	N/A	Set this parameter based on the network plan.

The following table describes the parameters that must be set to configure the reference relationship between the **gNBCUX2** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU X2 ID	**gNBCUX2**.***gNBCuX2Id***	Set this parameter based on the network plan.
Control Plane Endpoint Group ID	**gNBCUX2**.***CpEpGroupId***	Set this parameter based on the network plan.
User Plane Endpoint Group ID	**gNBCUX2**.***UpEpGroupId***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** and **gNBInterfaceParam** MOs.

Parameter Name	Parameter ID	Setting Notes
X2 SON Configuration Switch	**gNBX2SonConfig**.***X2SonConfigSwitch***	It is recommended that the **X2SON_X2U_AUTO_DEL_OPT_SWITCH** option be selected if X2-U link self-setup procedures need to be separated for different operators in NSA networking.
Interface Self-Confg Optimization Switch	**gNBInterfaceParam**.***InterfaceSelfCfgOptSw***	It is recommended that the **X2_UP_SELF_SETUP_NO_MOCN_SW** option be selected if X2-U link self-removal procedures need to be separated for different operators in NSA networking.

If the SON function is used to automatically configure, update, and remove S1 and X2 interfaces, the **GTRANSPARA**.***EPOBJAUTOCFGSW*** parameter must be set to its default value **ENABLE**.

4.8.1.2 Using MML Commands

Before using MML commands, refer to **4.7 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

X2 Self-Setup (Automatic Peer Configuration)

Configuring parameters on the eNodeB (IPv4/IPv6 single-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an endpoint group for the X2 interface (in IPv4 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an endpoint group for the X2 interface (in IPv6 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
```

```
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the eNodeB side
MOD GLOBALPROCSWITCH: InterfaceSetupPolicySw=LTE_NR_X2_SON_SETUP_SW-1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//(Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x", MTUSWITCH=OFF;
//New model
//(Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPROUTE4: RTIDX=0, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x",
MTUSWITCH=OFF;
//Turning on the GTP-U static check switches. (GEPMODELPARA.StaticChkMode is a global parameter that,
once modified, affects all types of interfaces configured on the base station.)
MOD EPGROUP: EPGROUPID=1, STATICCHK=ENABLE;
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
```

Configuring parameters on the eNodeB (IPv4/IPv6 dual-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422, SCTPTEMPLATEID=0;
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the eNodeB side
MOD GLOBALPROCSWITCH: InterfaceSetupPolicySw=LTE_NR_X2_SON_SETUP_SW-1;
```

Configuring parameters on the gNodeB (IPv4/IPv6 single-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an endpoint group for the X2 interface (in IPv4 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an endpoint group for the X2 interface (in IPv6 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
```



```
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the gNodeB side
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_SETUP_SWITCH-1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//Configuring a source IP route for each of the two local IP addresses of the X2 interface by using the local
IP address as the source IP address (The two IP addresses of the gNodeB are in the network segment of the
route.)
ADD SRCIPRT: SRCRTIDX=0, SN=6, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
//New model
//Configuring a source IP route for each of the two local IP addresses of the X2 interface by using the local
IP address as the source IP address (The two IP addresses of the gNodeB are in the network segment of the
route.)
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
ADD SRCIPROUTE4: SRCRTIDX=1, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
//Adding a control-plane host for the X2 interface (Two local IP addresses must be configured for the SCTP
host.)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2V4="x.x.x.x", SIGIP2SECSWITCH=DISABLE, PN=36422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF,
SCTPTEMPLATEID=0;
//Adding user-plane hosts for the X2 interface (Two user-plane hosts must be configured.)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=10, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
//Adding an endpoint group for the X2 interface: Turning on the GTP-U static check switches.
(GEPMODELPARA.StaticChkMode is a global parameter that, once modified, affects all types of interfaces
configured on the base station.)
MOD EPGROUP: EPGROUPID=1, STATICCHK=ENABLE;
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
//Adding the two X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=10;
```

Configuring parameters on the gNodeB (IPv4/IPv6 dual-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the gNodeB side
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_SETUP_SWITCH-1;
```


NOTE

For details about descriptions of MML-based configurations of the packet filtering protected X2 self-setup function, see related descriptions of "Activating ACL-based packet filtering (automatic ACL rule configuration)" in MML-based configurations of the integrated firewall function in *Equipment Security*.

X2 Self-Setup (Manual Peer Configuration)

Configuring parameters on the eNodeB (IPv4/IPv6 single-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane peer (in IPv4 networking)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;
//Adding an X2 control-plane peer (in IPv6 networking)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 user-plane peer to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding the X2 control-plane peer to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//(Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x", MTUSWITCH=OFF;
//New model
//(Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPROUTE4: RTIDX=0, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x",
MTUSWITCH=OFF;
//Adding a control-plane peer for the X2 interface (Two peer IP addresses must be configured for the SCTP
peer.)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP2V4="x.x.x.x", PN=36422;
//Adding user-plane peers for the X2 interface (Two user-plane peers must be configured.)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an endpoint group for the X2 interface: Turning on the GTP-U static check switches.
(GEPMODELPARA.StaticChkMode is a global parameter that, once modified, affects all types of interfaces
configured on the base station.)
MOD EPGROUP: EPGROUPID=1, STATICCHK=ENABLE;
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
```

```
//Adding the two X2 user-plane peers to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=2;
//Adding the SCTP peer of the X2 interface to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
```

Configuring parameters on the eNodeB (IPv4/IPv6 dual-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane peer through either of the following configurations
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;
ADD SCTPPEER: SCTPPEERID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422;
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer through either of the following configurations (The IP version on the user
plane must be the same as that on the control plane.)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
//Adding the X2 user-plane peers to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//Adding the X2 control-plane peers to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=2;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
```

Configuring parameters on the gNodeB (IPv4/IPv6 single-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane peer (in IPv4 networking)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;
//Adding an X2 control-plane peer (in IPv6 networking)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
```

```
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 user-plane peer to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding the X2 control-plane peer to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//For each of the two local IP addresses of the X2 interface, configuring an IP route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD SRCIPRT: SRCRTIDX=0, SN=6, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
//New model
//For each of the two local IP addresses of the X2 interface, configuring an IP route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
ADD SRCIPROUTE4: SRCRTIDX=1, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
//Adding a control-plane host for the X2 interface (Two local IP addresses must be configured for the SCTP
host.)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2V4="x.x.x.x", SIGIP2SECSWITCH=DISABLE, PN=36422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF,
SCTPTEMPLATEID=0;
//Adding user-plane hosts for the X2 interface (Two user-plane hosts must be configured.)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=10, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
//Adding an endpoint group for the X2 interface: Turning on the GTP-U static check switches.
(GEPMODELPARA.StaticChkMode is a global parameter that, once modified, affects all types of interfaces
configured on the base station.)
MOD EPGROUP: EPGROUPID=1, STATICCHK=ENABLE;
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
//Adding the two X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=10;
```

Configuring parameters on the gNodeB (IPv4/IPv6 dual-stack transmission)

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//(Required for manual X2 control-plane peer configuration) Adding an X2 control-plane peer through
either of the following configurations
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;
ADD SCTPPEER: SCTPPEERID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422;
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//(Required for manual X2 user-plane peer configuration) Adding an X2 user-plane peer through either of
the following configurations (It is recommended that the IP version on the user plane be the same as that
on the control plane.)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
```

```
//(Required for manual X2 user-plane peer configuration) Adding the X2 user-plane peer to the endpoint group through either of the following configurations
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//(Required for manual X2 control-plane peer configuration) Adding the X2 control-plane peer to the endpoint group through either of the following configurations
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=2;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
```

X2 Self-Removal

Configuring parameters on the eNodeB

```
//Activating X2 self-removal based on link faults
MOD ENODEBALGOEXTSWITCH: X2SonDelTimerForLNX2Fault=10080;
//Activating X2 self-removal based on the initial setup state
MOD ENODEBALGOEXTSWITCH: GnbX2InitFailDelSwitch=ON;
//Activating X2 self-removal based on X2 usage
MOD ENODEBALGOEXTSWITCH: X2SonDelTimerForLNX2Usage=10080, X2SonDeleteGnbAddCntThld=0;
```

Configuring parameters on the gNodeB

```
//Activating X2 self-removal based on link faults
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_DEL_FOR_X2FAULT_SWITCH-1, X2SonDeleteTimerForX2Fault=10080;
//Activating X2 self-removal triggered by immediate faults after initial X2 self-setup
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2INIT_FAIL_DEL_SWITCH-1;
//Activating X2 self-removal based on X2 usage
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_DEL_FOR_X2USAGE_SWITCH-1, X2SonDeleteGnbAddCntThld=0, X2SonDeleteTimerForX2Usage=10080;
```

X2 Blacklist and Whitelist

Configuring parameters on the eNodeB

```
//Adding a peer gNodeB to the blacklist
ADD ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=1, ListType=BLACKLIST;
//Adding a peer gNodeB to the whitelist
ADD ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=2, ListType=WHITELIST;
//Removing peer gNodeBs from the blacklist/whitelist
RMV ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=1;
RMV ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=2;
```

Other Optimization Functions

Run the following commands to separate X2-U link self-setup/self-removal procedures for different operators in NSA networking.

```
MOD ENODEBALGOEXTSWITCH: NetworkPrfmOptSwitch=SHARE_FREQ_X2_UP_OPT_SW-1;
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_X2U_AUTO_DEL_OPT_SWITCH-1;
MOD GNBINTERFACEPARAM: InterfaceSelfCfgOptSw=X2_UP_SELF_SETUP_NO_MOCN_SW-1;
```

4.8.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-**

Deployment Guidelines > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.8.2 Activation Verification

The following table lists the counters on the LTE side.

Counter ID	Counter Name
1526727202	**L.Sig.X2.SendSetup.Att**
1526727203	**L.Sig.X2.SendSetup.Succ**
1526728766	**L.X2.Unavail.Dur.Sys**
1526745966	**L.TRPIP.X2UTxMeanSpeed**
1526745967	**L.TRPIP.X2UTxMaxSpeed**
1526745968	**L.TRPIP.X2URxMeanSpeed**
1526745969	**L.TRPIP.X2URxMaxSpeed**
1526745960	**L.TRPIP.X2UTxBytes**
1526745961	**L.TRPIP.X2URxBytes**
1526729660	**L.Signal.Num.X2**

The following table lists the counters on the NR side.

Counter ID	Counter Name
1911816492	**N.TRPIP.gNB.X2UTxMeanSpeed**
1911816493	**N.TRPIP.gNB.X2UTxMaxSpeed**
1911816494	**N.TRPIP.gNB.X2URxMeanSpeed**
1911816495	**N.TRPIP.gNB.X2URxMaxSpeed**
1911816496	**N.TRPIP.gNB.X2UTxBytes**
1911816497	**N.TRPIP.gNB.X2URxBytes**

4.8.3 Network Monitoring

None

5 S1-U Self-Management

5.1 S1-U Self-Setup

5.1.1 Overview

In EN-DC, the S1-U interface can be configured only in endpoint mode for an eNodeB or a gNodeB. This section describes only the gNodeB S1-U configuration. For details about eNodeB's support for configuring the S1-U interface in endpoint mode, see *S1 and X2 Self-Management* in *eRAN Feature Documentation*.

The S1-U host must be configured manually. For details, see **5.5.1.2 Using MML Commands**. The S1-U peer can be configured manually or automatically.

- Manually

  The **USERPLANEPEER** MO must be configured for the peers manually. If a gNodeB has multiple S1-U interfaces, multiple peers must be manually added.

- Automatically

  The peers do not need to be configured manually. When a DC service is required and the gNodeB detects that there are no operational S1-U paths, the gNodeB uses the S-GW IP address sent by the eNodeB as the user-plane IP address of the peer.

NOTE

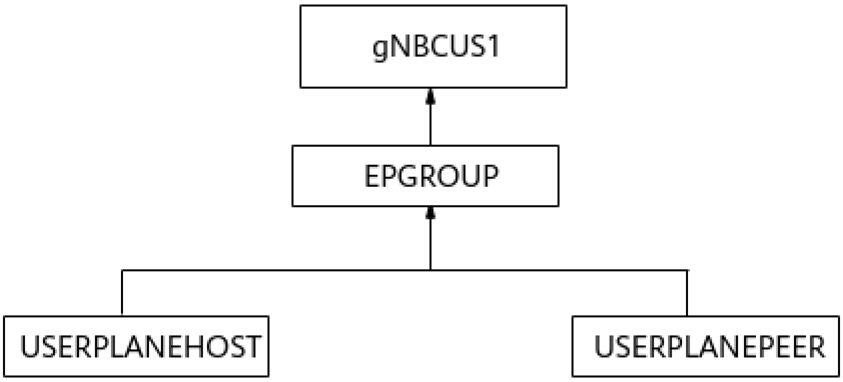
For S1-U self-setup with peers automatically configured in an eNodeB or a gNodeB, two user-plane peers (**USERPLANEPEER** MOs) can be automatically created at the same time. If there are self-setup requests from more than two user-plane peers at the same time, excess requests will be rejected, which may result in UE access failures.

5.1.2 IPv4/IPv6 Single-Stack Transmission

5.1.2.1 S1-U Self-Setup with Peers Manually Configured

Figure 5-1 shows the relationships between MOs during S1-U self-setup with peers manually configured.

Figure 5-1 Relationships between MOs during S1-U self-setup with peers manually configured



The details are as follows:

1. The following MOs are configured: **USERPLANEPEER**, **USERPLANEHOST**, **EPGROUP**, and **gNBCUS1**. There are two configuration requirements:
   a. The **USERPLANEHOST** and **USERPLANEPEER** MOs are added to the **EPGROUP** MO by running the **ADD UPHOST2EPGRP** and **ADD UPPEER2EPGRP** commands, respectively.
   b. The **EPGROUP** MO is referenced by the **gNBCUS1** MO.
2. The gNodeB automatically sets up an S1-U transmission link using **USERPLANEHOST**.***LOCIPV4*** (the local S1 service IP address) and **USERPLANEPEER**.***PEERIPV4*** (the peer S-GW IP address).
3. When a UE initiates a DC service, the eNodeB sends an SgNB Addition Request to the gNodeB.
4. The eNodeB sends the gNodeB's IP address to the core network through the E-RAB Modification Indication procedure. An S1-U interface is set up on the core network side as specified in 3GPP TS 37.340.

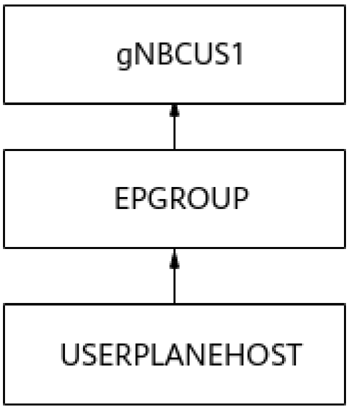
NOTE

- If the IP address in the manually configured **USERPLANEPEER** MO is different from the IP address of the S-GW carried in the SgNB Addition Request message, the peer can be automatically configured based on the S-GW IP address. For details, see **2**.
- S1-U transmission links in the same endpoint group must be available to all operators.

5.1.2.2 S1-U Self-Setup with Peers Automatically Configured

Figure 5-2 shows the relationships between MOs during S1-U self-setup with peers automatically configured.

Figure 5-2 Relationships between MOs during S1-U self-setup with peers (**USERPLANEPEER** MO) automatically configured



The S1-U transmission link self-setup process is as follows:

1. When a UE initiates a DC service, the eNodeB sends an SgNB Addition Request message carrying the S-GW IP address to the gNodeB.
2. After the request is accepted, the gNodeB sends an SgNB Addition Request Acknowledge message carrying the IP address of the gNodeB S1-U interface to the eNodeB.
   a. The gNodeB automatically generates a **USERPLANEPEER** MO based on the S-GW IP address and adds the **USERPLANEPEER** MO to the **EPGROUP** MO.
   b. The gNodeB sets up an S1-U transmission link based on the manually configured **USERPLANEHOST** and **EPGROUP** MOs and the automatically generated **USERPLANEPEER** MO.
3. The eNodeB sends the gNodeB's IP address to the core network through the E-RAB Modification Indication procedure. An S1-U interface is set up on the core network side as specified in 3GPP TS 37.340.

5.1.2.3 Packet Filtering Protected S1-U Self-Setup

The procedures for packet filtering protected S1-U self-setup are similar to those described in **4.1.2.3 Packet Filtering Protected X2 Self-Setup**, except for the differences in X2 and S1-U interfaces.

After ACL rule self-configuration for packet filtering is enabled for the S1-U interface in endpoint mode, the base station automatically generates ACL rules that completely match the IP addresses of the user-plane host and peer. Subsequent packet filtering is performed based on these rules. You are advised not to manually configure ACL rules for the S1 interface again. Otherwise, ACL rule capacity will decrease.

NOTE

When manually configuring an **ACLRULE** MO on an eNodeB or gNodeB for an S-GW, ensure that the **ACLRULE.*ACLID*** parameter is set to the same value as the **PACKETFILTER.*ACLID*** (old model) or **PACKETFILTERING.*ACLID*** (new model) for packet filtering. Otherwise, transmission links will fail. (When the **GTRANSPARA**.***TRANSCFGMODE*** parameter is set to **OLD**, the old model is used. When this parameter is set to **NEW**, the new model is used.)

5.1.3 IPv4/IPv6 Dual-Stack Transmission

The S1-U interface supports IPv4, IPv6, or IPv4/IPv6 dual-stack transmission. An IPv4 or IPv6 S1-U link will be established based on the IP version configured on the gNodeB.

- If the S1-U host is configured to use IPv4, the eNodeB sends an SgNB Addition Request message carrying the S-GW IPv4 address to the gNodeB. Then, the gNodeB responds with an SgNB Addition Request Acknowledge message carrying its S1-U IPv4 address to the eNodeB. An IPv4 S1-U link will then be established.
- If the S1-U host is configured to use IPv6, the eNodeB sends an SgNB Addition Request message carrying the S-GW IPv6 address to the gNodeB. Then, the gNodeB responds with an SgNB Addition Request Acknowledge message carrying its S1-U IPv6 address to the eNodeB. An IPv6 S1-U link will then be established.
- If the S1-U host is configured to use IPv4/IPv6 dual-stack and **EPGROUP.*IPVERPREFERENCE (5G gNodeB, LTE eNodeB)*** is set to **IPv4**, the eNodeB sends an SgNB Addition Request message carrying the S-GW IPv4 address to the gNodeB. Then, the gNodeB responds with an SgNB Addition Request Acknowledge message carrying its S1-U IPv4 address to the eNodeB. An IPv4 S1-U link will then be established.
- If the S1-U host is configured to use IPv4/IPv6 dual-stack and **EPGROUP.*IPVERPREFERENCE (5G gNodeB, LTE eNodeB)*** is set to **IPv6**, the eNodeB sends an SgNB Addition Request message carrying the S-GW IPv6 address to the gNodeB. Then, the gNodeB responds with an SgNB Addition Request Acknowledge message carrying its S1-U IPv6 address to the eNodeB. An IPv6 S1-U link will then be established.

NOTE

A change in the **EPGROUP.*IPVERPREFERENCE (5G gNodeB, LTE eNodeB)*** parameter setting will not cause the S1-C link or S1-U link to reestablish. The parameter setting will not be checked until a new S1-U link is established or an S1-U link is reestablished.

5.2 S1-U Self-Removal

When the **USERPLANEPEER**.***CTRLMODE*** parameter is set to **AUTO_MODE** on the base station side, the base station can automatically remove the S1-U transmission links and the **USERPLANEPEER** MOs. When this parameter is set to **MANUAL_MODE**, the base station does not automatically remove the S1-U transmission links and the **USERPLANEPEER** MOs.

When the **GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)*** parameter is set to **ENABLE**, the gNodeB determines whether to perform S1-U self-removal in any of the following situations:

- Self-removal based on link faults

  If the GTP-U detection result indicates that all the transmission links corresponding to **USERPLANEPEER** MOs are faulty, the gNodeB periodically (every 60 minutes) attempts to remove those transmission links (up to 50 links at one time) that remain faulty for over one hour and the corresponding **USERPLANEPEER** MOs.

- Self-removal based on the aging mechanism
  - In case of an excess over the specifications of a base station

    The gNodeB marks a transmission link as aged and processes as follows if the link is not used by any service.

    - The base station performs self-removal based on the aging mechanism once every hour and checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the base station removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs. If it does not, no aged transmission links are removed.
    - When new transmission links need to be established, the base station checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the base station removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs, and establishes new transmission links. If it does not, no aged transmission links are removed.
    - If the maximum allowed number of transmission links is reached and there is no aged link, the gNodeB rejects new UEs' access requests.
  - In case of an excess over the specifications of a single endpoint group

    "Self-removal in case of an excess over the specifications of a single endpoint group" takes effect for the eNodeB or gNodeB when the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (5G gNodeB, LTE eNodeB)*** parameter is selected. Self-removal based on the aging mechanism starts when the number of peers in a single endpoint group exceeds 90% of the maximum number allowed by the in-use board. The condition for marking a link as aged and the self-removal mechanism are the same as those for self-removal in case of an excess over the specifications of a base station.

When GTP-U static check is enabled (by setting the **USERPLANEPEER**.***STATICCHK*** parameter), ALM-25954 User Plane Fault is reported when an S1-U peer fails to communicate with any S1-U host in the endpoint group to which the S1-U peer belongs. In this case, the S1-U peer is considered invalid.

5.3 Network Analysis

5.3.1 Benefits

S1-U self-setup simplifies configuration operations and reduces costs for operators.

5.3.2 Impacts

The impacts between this function and other functions are described in **5.4.2.3 Function Impacts**.

This function has no impact on the network.

5.4 Requirements

5.4.1 Licenses

None

5.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Automatic user plane removal	**GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)***	*X2 and S1 Self-Management in NSA Networking*	Ensure that the **GEPMODELPARA**.***UPAUTODELSW (LTE eNodeB, 5G gNodeB)*** parameter is set to **ENABLE** before enabling "self-removal in case of an excess over the specifications of a single endpoint group" (by selecting the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)*** parameter).

5.4.2.2 Mutually Exclusive Functions

None

5.4.2.3 Function Impacts

None

5.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For LTE, a UMPT is configured as the main control board.

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

Currently, only the UMPT supports user-plane load sharing.

RF Modules

N/A

5.4.4 Others

None

5.5 Operation and Maintenance

5.5.1 Data Configuration

5.5.1.1 Data Preparation

For details about basic configurations on the user plane, see **Configuring Common Parameters** in **4.8.1.1 Data Preparation**.

Parameters to be configured on the gNodeB side:

(Optional) When **GTRANSPARA.*TRANSCFGMODE*** is set to **OLD** and user-plane load sharing is required, set the following parameters in an **SRCIPRT** MO to configure a source IP route.

Parameter Name	Parameter ID	Setting Notes
Source Route Index	**SRCIPRT.*SRCRTIDX***	Set this parameter based on the network plan.
Source IP Address	**SRCIPRT.*SRCIP***	This parameter specifies the local IP address of the base station. It is the local IP address of the S1-U interface.
Subboard Type	**SRCIPRT.*SBT***	Set this parameter to **BASE_BOARD(Base Board)**.
Route Type	**SRCIPRT.*RTTYPE***	Set this parameter to **NEXTHOP(Next Hop)** in Ethernet networking.
Interface Type	**SRCIPRT.*IFT***	This parameter specifies the type of the port.
Next Hop IP	**SRCIPRT.*NEXTHOP***	• This parameter is valid only when the **IPRT.*RTTYPE (LTE eNodeB, 5G gNodeB)*** parameter is set to **NEXTHOP(Next Hop)**. • Generally, based on the network plan, this parameter is set to the IP address of the gateway on the transmission network connecting to the base station.

Parameter Name	Parameter ID	Setting Notes
Priority	**SRCIPRT.*PREF***	This parameter needs to be configured if IP route backup is required. The priority is used for route identification. A smaller parameter value indicates a higher priority. Generally, the route with a higher priority is activated. The base station does not support route load sharing. Routes to the same destination network segment must have different priorities.

(Optional) When **GTRANSPARA.*TRANSCFGMODE*** is set to **NEW** and user-plane load sharing is required, set the following parameters in an **SRCIPROUTE4** MO to configure a source IP route.

Parameter Name	Parameter ID	Setting Notes
Source Route Index	**SRCIPROUTE4.*SRCRTIDX***	Set this parameter based on the network plan.
Source IP Address	**SRCIPROUTE4.*SRCIP***	This parameter specifies the local IP address of the base station. It is the local IP address of the S1-U interface.
Route Type	**SRCIPROUTE4.*RTTYPE***	Set this parameter to **NEXTHOP(Next Hop)** in Ethernet networking.
Port Type	**SRCIPROUTE4.*PT***	This parameter specifies the port type of a source IPv4 route.
Port ID	**SRCIPROUTE4.*PORTID***	This parameter specifies the port ID of a source IPv4 route.

Parameter Name	Parameter ID	Setting Notes
Next Hop IP	**SRCIPROUTE4**.***NEXTHOP***	This parameter specifies the next-hop IP address of a source IPv4 route. • This parameter is valid only when the **IPRT**.***RTTYPE (LTE eNodeB, 5G gNodeB)*** parameter is set to **NEXTHOP(Next Hop)**. • Generally, based on the network plan, this parameter is set to the IP address of the gateway on the transmission network connecting to the base station.
Priority	**SRCIPROUTE4**.***PREF***	This parameter needs to be configured if IP route backup is required. The priority is used for route identification. A smaller parameter value indicates a higher priority. Generally, the route with a higher priority is activated. The base station does not support route load sharing. Routes to the same destination network segment must have different priorities.

(Optional) Set the following parameters in a **GTPU** MO when active/standby user plane is required.

Parameter Name	Parameter ID	Setting Notes
Static Check Switch	**GTPU**.***STATICCHK***	This parameter specifies whether to enable the GTP-U static check function. When active/standby user plane is required, set this parameter to **ENABLE**, indicating that the GTP-U static check is enabled.

(Optional) Set the following parameters in a **GEPMODELPARA** MO when active/standby user plane is required.

Parameter Name	Parameter ID	Setting Notes
Static Check Mode	**GEPMODELPARA**.***STATICCHKMODE***	This parameter specifies the effective mode of the GTP-U static check switch in endpoint mode. When active/standby user plane is required, set this parameter to **EPSTATICCHK**, indicating that the endpoint group static check switch takes effect. This parameter is a global parameter that, once modified, affects all types of interfaces configured on the base station.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	**USERPLANEHOST**.***UPHOSTID***	N/A
IP Version	**USERPLANEHOST**.***IPVERSION***	● If the peer uses IPv4, set this parameter to **IPv4**. ● If the peer uses IPv6, set this parameter to **IPv6**.
Local IP Address	**USERPLANEHOST**.***LOCIPV4***	This parameter specifies the IP address of the user-plane host.
Local IPv6 Address	**USERPLANEHOST**.***LOCIPV6***	This parameter specifies the IPv6 address of the user-plane host.

(Optional, required for manual peer configuration) The following table describes the parameters that must be set in a **USERPLANEPEER** MO.

Parameter Name	Parameter ID	Setting Notes
User Plane Peer ID	**USERPLANEPEER**.***UPPEERID***	N/A

Parameter Name	Parameter ID	Setting Notes
IP Version	**USERPLANEPEER**.***IPVERSION***	• If the peer uses IPv4, set this parameter to **IPv4**. • If the peer uses IPv6, set this parameter to **IPv6**.
Peer IP Address	**USERPLANEPEER**.***PEERIPV4***	Set this parameter based on the network plan.
Peer IPv6 Address	**USERPLANEPEER**.***PEERIPV6***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	**EPGROUP**.***EPGROUPID***	N/A
User Plane Host List	**EPGROUP**.***USERPLANEHOSTLIST***	N/A
User Plane Peer List	**EPGROUP**.***USERPLANEPEERLIST***	N/A
IP Protocol Version Preference	**EPGROUP**.***IPVERPREFERENCE***	This parameter specifies the preferred IP version for automatic link setup when both IPv4 and IPv6 addresses are configured on the local and peer ends.
Packet Filter ACL Rule Auto-Setup-Deletion SW	**EPGROUP**.***PACKETFILTERSWITCH***	This parameter specifies whether to enable the ACL rule self-configuration function for packet filtering.
IPv6 Packet Filter ACL Rule Auto-Setup-Deletion SW	**EPGROUP**.***PACKETFILTERSWITCH6***	This parameter specifies whether to enable the ACL rule self-configuration function for IPv6 packet filtering.

The following table describes the parameters that must be set to configure the reference relationship between the **gNBCUS1** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU S1 ID	**gNBCUS1**.*gNBCuS1Id*	N/A
User Plane Endpoint Group ID	**gNBCUS1**.*UpEpGroupId*	N/A

If the SON function is used to automatically configure, update, and remove S1 and X2 interfaces, the **GTRANSPARA**.*EPOBJAUTOCFGSW* parameter must be set to its default value **ENABLE**.

5.5.1.2 Using MML Commands

Before using MML commands, refer to 5.4 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

S1-U Self-Setup (Automatic Peer Configuration) – IPv4/IPv6 Single-Stack Configuration

- When user-plane load sharing and active/standby user plane are not required:

```
//Adding an S1-U host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an S1-U host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//Adding an endpoint group (in IPv4 networking)
ADD EPGROUP: EPGROUPID=1, PACKETFILTERSWITCH=ENABLE;
//Adding an endpoint group (in IPv6 networking)
ADD EPGROUP: EPGROUPID=1, PACKETFILTERSWITCH6=ENABLE;
//Adding the S1-U host to the endpoint group (in IPv4 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//Adding the S1-U host to the endpoint group (in IPv6 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

- When user-plane load sharing is required:

```
//Adding a source IP route when GTRANSPARA.TRANSCFGMODE is set to OLD
ADD SRCIPRT: SRCRTIDX=0, SN=7, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x", PREF=60;
//Adding a source IP route when GTRANSPARA.TRANSCFGMODE is set to NEW
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x", PREF=60;
//Adding S1-U hosts
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0;
//Adding the S1-U hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

- When active/standby user plane is required:

```
//Enabling the GTP-U static check function. (GEPMODELPARA.StaticChkMode is a global parameter that, once modified, affects all types of interfaces configured on the base station.)
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
//Adding S1-U hosts (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=SLAVE;
//Adding S1-U hosts (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE, FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE, FLAG=SLAVE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the S1-U hosts to the endpoint group (in IPv4 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//Adding the S1-U hosts to the endpoint group (in IPv6 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=2;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=3;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

S1-U Self-Setup (Automatic Peer Configuration) – IPv4/IPv6 Dual-Stack Configuration

- When user-plane load sharing and active/standby user plane are not required:

```
//Adding S1-U hosts
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0, IPVERPREFERENCE=IPv6, PACKETFILTERSWITCH6=ENABLE;
//Adding the S1-U hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

- When user-plane load sharing is required:
  - Only one IP version can be used for user-plane data backup, and IPv4- and IPv6-based data backup is not supported at the same time.
  - The user-plane data backup configuration in a single endpoint group when the IPv4/IPv6 dual-stack is used is the same as that when a single stack is used.
- When active/standby user plane is required:
  - If there is a user-plane host with **USERPLANEHOST**.***FLAG (5G gNodeB, LTE eNodeB)*** set to **SLAVE** in the endpoint group, the **USERPLANEHOST**.***IPVERSION (5G gNodeB, LTE eNodeB)*** parameter of all user-plane hosts referenced by this endpoint group must be set to **IPv4** or **IPv6**. Only one IP version can be used for user-plane data backup, and IPv4- and IPv6-based data backup is not supported at the same time.
  - The user-plane data backup configuration in a single endpoint group when the IPv4/IPv6 dual-stack is used is the same as that when a single stack is used.

S1-U Self-Setup (Manual Peer Configuration) – Single-Stack Configuration

- When user-plane load sharing and active/standby user plane are not required:

```
//Adding an S1-U host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an S1-U host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//Adding an S1-U peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an S1-U peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//Adding an endpoint group (in IPv4 networking)
ADD EPGROUP: EPGROUPID=0, PACKETFILTERSWITCH=ENABLE;
//Adding an endpoint group (in IPv6 networking)
ADD EPGROUP: EPGROUPID=0, PACKETFILTERSWITCH6=ENABLE;
//Adding the S1-U host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//(Required only for manual S1-U peer configuration in IPv4 networking) Adding the S1-U peer to the
endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
//(Required only for manual S1-U peer configuration in IPv6 networking) Adding the S1-U peer to the
endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

- When user-plane load sharing is required:

```
//Adding S1-U hosts (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding S1-U hosts (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//Adding an S1-U peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an S1-U peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0;
//Adding the S1-U hosts to the endpoint group (in IPv4 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//Adding the S1-U hosts to the endpoint group (in IPv6 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=2;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=3;
//(Required only for manual S1-U peer configuration in IPv4 networking) Adding the S1-U peer to the
endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
//(Required only for manual S1-U peer configuration in IPv6 networking) Adding the S1-U peer to the
endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

- When active/standby user plane is required:

```
//Enabling the GTP-U static check function. (GEPMODELPARA.StaticChkMode is a global parameter
that, once modified, affects all types of interfaces configured on the base station.)
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
//Adding S1-U hosts (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=SLAVE;
//Adding S1-U hosts (in IPv6 networking)
```

```
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE, FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE, FLAG=SLAVE;
//Adding an S1-U peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an S1-U peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the S1-U hosts to the endpoint group (in IPv4 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//Adding the S1-U hosts to the endpoint group (in IPv6 networking)
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=2;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=3;
//(Required only for manual S1-U peer configuration in IPv4 networking) Adding the S1-U peer to the
endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
//(Required only for manual S1-U peer configuration in IPv6 networking) Adding the S1-U peer to the
endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

S1-U Self-Setup (Manual Peer Configuration) – Dual-Stack Configuration

- When user-plane load sharing and active/standby user plane are not required:

```
//Adding S1-U hosts
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//(Required only for manual S1-U peer configuration) Adding S1-U peers
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx",
IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0, IPVERPREFERENCE=IPv6, PACKETFILTERSWITCH6=ENABLE;
//Adding the S1-U hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
//(Required only for manual S1-U peer configuration) Adding the S1-U peers to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=1;
//Adding an S1-U object
ADD GNBCUS1: gNBCuS1Id=0, UpEpGroupId=0;
```

- If user-plane load sharing is required, the user-plane load sharing configuration in a single endpoint group when the dual-stack is used is the same as that when a single stack is used.
- If active/standby user plane is used, the user-plane data backup configuration in a single endpoint group when the dual-stack is used is the same as that when a single stack is used.

5.5.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.2 Activation Verification

The following table lists the related counters.

Counter ID	Counter Name
1911816504	**N.TRPIP.gNB.S1UTxMeanSpeed**
1911816505	**N.TRPIP.gNB.S1UTxMaxSpeed**
1911816506	**N.TRPIP.gNB.S1URxMeanSpeed**
1911816507	**N.TRPIP.gNB.S1URxMaxSpeed**
1911816508	**N.TRPIP.gNB.S1UTxBytes**
1911816509	**N.TRPIP.gNB.S1URxBytes**

5.5.3 Network Monitoring

None

6 Direct IPsec for X2 Interface

6.1 Principles

The X2 interface delay on the live network is affected by the deployment location of the SeGW and network topology. The delay for data transmission is long. Direct IPsec is introduced to set up a direct IPsec tunnel between a gNodeB and an eNodeB, thereby reducing the X2 interface delay.

Figure 6-1 and **Figure 6-2** show the typical networking of direct IPsec for the X2 interface between a gNodeB and an eNodeB.

Figure 6-1 Typical networking 1 of direct IPsec

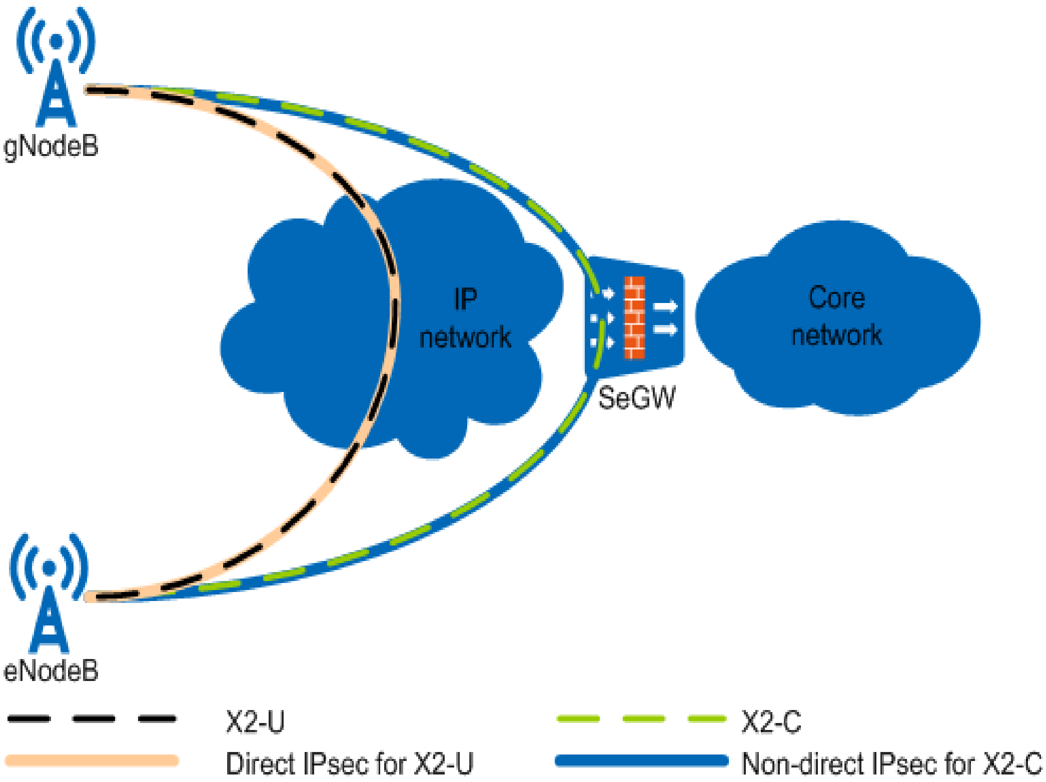
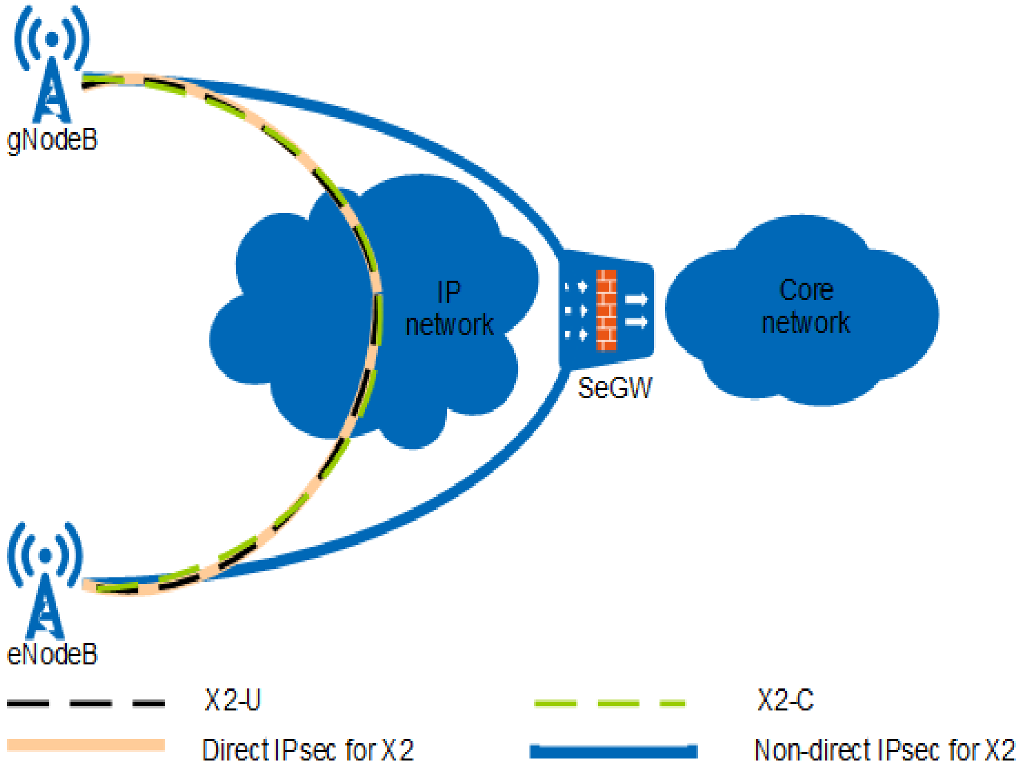


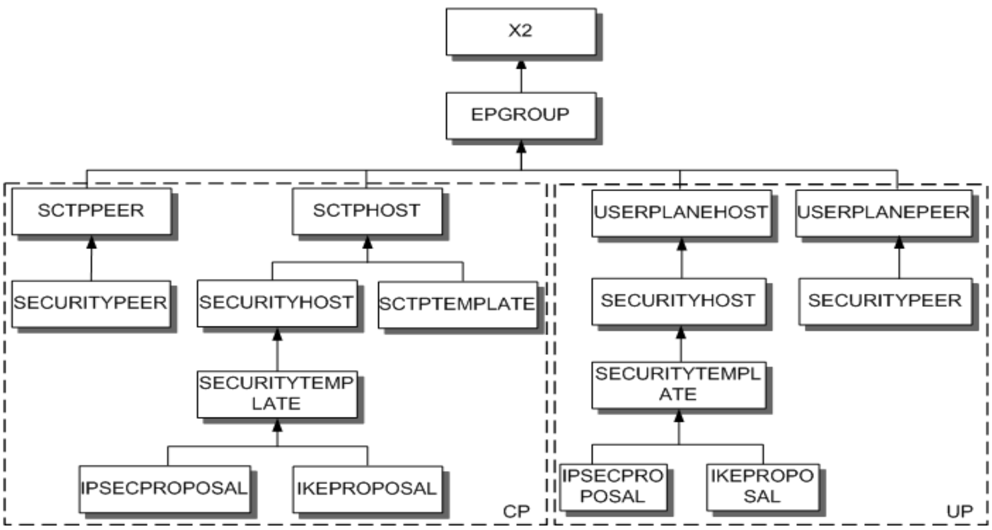
Figure 6-2 Typical networking 2 of direct IPsec



6.1.1 Direct IPsec Self-Setup

Direct IPsec tunnels can be set up between an eNodeB and a gNodeB through automatic or manual configuration of X2 control-plane and user-plane peers (corresponding to **SCTPPEER** and **USERPLANEPEER** MOs). If the **SCTPPEER** or **USERPLANEPEER** MO is manually configured, the **SECURITYPEER** MO must be configured as well. **Figure 6-3** shows the configuration model.

Figure 6-3 Configuration model for direct IPsec tunnel setup between an eNodeB and a gNodeB



NOTE

In IPv6 scenarios, the **SECURITYPEER** MO cannot be manually configured.

Direct IPsec is classified into direct IPv4 IPsec and direct IPv6 IPsec.

- For each X2 interface, only one security host IPv4 address can be configured for direct IPv4 IPsec self-setup, and only one security host IPv6 address can be configured for direct IPv6 IPsec self-setup.
- In IPv4/IPv6 dual-stack scenarios, only one security host IPv4 address and one security host IPv6 address can be configured for direct IPsec self-setup.

If each of the source and target base stations is configured with IPv4 and IPv6 addresses and the source base station is enabled with direct IPv4 IPsec self-setup and direct IPv6 IPsec self-setup, then the target base station selects an IP version for direct IPsec tunnel setup based on the value of the **EPGROUP**.***IPVERPREFERENCE (LTE eNodeB, 5G gNodeB)*** parameter.

6.1.1.1 Direct IPv4 IPsec Self-Setup

Automatic Configurations

IPv4 IPsec supports two configuration modes: Ethernet-interface-based IPsec and tunnel-interface-based IPsec. The **IKECFG**.***IPSECCFGMODE*** parameter specifies the configuration mode to use. If this parameter is set to **ETHINTERFACE_BASED**, Ethernet-interface-based IPsec is used. If this parameter is set to **TUNNELITF_BASED**, tunnel-interface-based IPsec is used. The "tunnel-interface-based IPsec" configuration mode is supported only in LTE, NR, and multimode scenarios. No matter which IPv4 IPsec configuration mode is used, direct IPv4 IPsec self-setup for the X2 interface can be implemented using either of the following methods:

- Method 1: Direct IPv4 IPsec self-setup only for the X2 user plane
- Method 2: Direct IPv4 IPsec self-setup for the X2 control plane and user plane

NOTE

- Inter-base-station coordination-based services in non-ideal backhaul scenarios have latency requirements only for the X2 user plane. In addition, when the IPv4 IPsec configuration mode is Ethernet-interface-based IPsec, the IPv4 IPsec specification (number of IPv4 IPsec tunnels) of the eNodeB does not support direct IPv4 IPsec tunnels for both control and user planes of all X2 interfaces. Therefore, method 1 is recommended when the "Ethernet-interface-based IPsec" configuration mode is used.
- When the IPv4 IPsec configuration mode is tunnel-interface-based IPsec and the X2 control plane and user plane share an IP address, only method 2 is supported.
- It is recommended that the base stations at both ends of a direct IPv4 IPsec tunnel be configured with the same IPsec configuration mode, that is, either Ethernet-interface-based or tunnel-interface-based.

The following describes the two methods.

- Direct IPv4 IPsec self-setup only for the X2 user plane:

  The gNodeB/eNodeB uses the security IP address carried in the X2 self-setup message as the IP address of the security peer (**IKEPEER** MO) of the X2 user plane. All IPv4 IPsec-related MOs need to be manually configured for the X2 control plane to set up an IPv4 IPsec tunnel between the gNodeB/eNodeB and SeGW.

In this method, the **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameter must be set to **ENABLE** and the **SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)*** parameter must be set to **DISABLE**. Otherwise, direct IPv4 IPsec self-setup cannot be achieved for the user plane.

The gNodeB/eNodeB sets up X2-C and X2-U transmission links based on the **X2**/**gNBCUX2**, **SCTPHOST**, **USERPLANEHOST**, and **EPGROUP** MOs, as well as the automatically generated **SCTPPEER** and **USERPLANEPEER** MOs.

The gNodeB/eNodeB sets up a direct IPv4 IPsec tunnel for the X2-U interface after automatically generating the **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model) MOs based on the preceding IPsec parameters, the IP address of the security peer included in the signaling messages, and the peer's control and user plane IP addresses. When **GTRANSPARA**.***TRANSCFGMODE (5G gNodeB, LTE eNodeB)*** is set to **OLD**, the old model is used. When this parameter is set to **NEW**, the new model is used. If the IPv4 IPsec configuration mode is tunnel-interface-based IPsec, only the new model can be used.

- Direct IPv4 IPsec self-setup for the X2 control plane and user plane:

  The base station uses the security IP address carried in the X2 self-setup message as the IP address of the security peers of the X2 control and user planes. Direct IPv4 IPsec is configured for both the X2 control plane and user plane, and the two planes must use the same security host (**SECURITYHOST** MO).

  In this method, the **SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)*** and **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameters must be set to **ENABLE**. Otherwise, direct IPv4 IPsec self-setup cannot be achieved for the control and user planes.

  The gNodeB/eNodeB sets up X2-C and X2-U transmission links based on the **X2**/**gNBCUX2**, **SCTPHOST**, **USERPLANEHOST**, and **EPGROUP** MOs, as well as the automatically generated **SCTPPEER** and **USERPLANEPEER** MOs.

  The gNodeB/eNodeB sets up direct IPv4 IPsec tunnels for the X2-C and X2-U interfaces after automatically generating the **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model) MOs based on the preceding IPsec parameters, the IP address of the security peer included in the signaling messages, and the peer's control and user plane IP addresses.

When X2 interface data matches multiple IPv4 IPsec tunnels, the **GTRANSPARA**.***DIRECTIPSECPRIMATCHSW (5G gNodeB, LTE eNodeB)*** parameter can be used to adjust the IPv4 IPsec policy matching priority, enabling the X2 interface to preferentially match direct IPv4 IPsec tunnels or IPsec tunnels connected to the SeGW.

- When this parameter is set to **DISABLE**, the IPv4 IPsec policy with a smaller sequence number (specified by **IPSECPOLICY**.***SPSN (5G gNodeB, LTE eNodeB)***) is preferentially matched.
- When this parameter is set to **ENABLE**, the IPv4 IPsec policy of a self-setup direct IPsec tunnel is preferentially matched. If the direct IPsec tunnel corresponding to the IPv4 IPsec policy is faulty, the IPv4 IPsec policy of a normal IPsec tunnel is preferentially matched.

Application Scenarios

- In the following scenarios, the IPv4 IPsec specification is insufficient, causing X2 self-setup failures:
  - When direct IPv4 IPsec self-setup is configured for both X2 control plane and user plane, a direct IPv4 IPsec tunnel needs to be set up for each of the control plane and user plane. However, the current IPv4 IPsec specification does not allow direct IPv4 IPsec tunnels for all X2 interfaces.
  - Assume that the IPsec specification cannot meet the service requirements. When the **GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **STRICT_MODE**, a manually configured non-direct-IPsec X2 interface will be removed due to a limitation on the direct IPv4 IPsec specification after it changes from non-direct-IPv4-IPsec mode to direct-IPv4-IPsec mode. When the **GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **LOOSE_MODE**, an X2 interface is automatically generated even if direct IPv4 IPsec setup fails.
- If direct IPv4 IPsec tunnels are required between the eNodeB and gNodeB of a co-MPT base station, it is recommended that the eNodeB and gNodeB use the same security host (**SECURITYHOST** MO).
- For the eNodeB and gNodeB of a co-MPT base station, the **SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)*** and **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameters of the eNodeB must be set to the same values as those of the gNodeB, respectively. Otherwise, the X2 interface may fail to be set up.
- When **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** is set to **TUNNELITF_BASED** and the control-plane and user-plane hosts of the X2 interface share an IP address, the **SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)*** and **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameters must be both set to **ENABLE** or **DISABLE**.

6.1.1.2 Direct IPv6 IPsec Self-Setup

In direct IPv6 IPsec self-setup for the X2 interface, direct IPv6 IPsec must be automatically set up for both control and user planes.

To allow direct IPv6 IPsec to be configured for both the X2 control plane and user plane, the two planes must use the same security host (**SECURITYHOST** MO). The base station sets up X2-C and X2-U transmission links based on the following MOs: **X2**/**gNBCUX2**, **SCTPHOST**, **USERPLANEHOST**, **EPGROUP**, as well as the automatically generated **SCTPPEER** and **USERPLANEPEER**. The local base station sets up direct IPv6 IPsec tunnels for the X2-C and X2-U interfaces based on the peer base station's information carried in the signaling messages and the automatically generated MOs. The peer base station's information includes its control-plane IPv6 IPsec address, user-plane IPv6 IPsec address, control-plane address, and user-plane address. The automatically generated MOs include **SECURITYPEER**, **ACL6**, **ACLRULE6**, **IPSECPOLICY**, **IKEPEER**, **IPSECBINDITF**, **TUNNELITF**, **INTERFACE**, and **POLICYBASEDROUTING6** (internal object).

The IPsec self-setup switches on the user and control planes (specified by **USERPLANEHOST**.***IPSECSWITCH*** and **SCTPHOST**.***SIGIP1SECSWITCH***, respectively) must be turned on for base stations on both ends of the X2 interface where direct IPv6 IPsec self-setup is required. Otherwise, direct IPv6 IPsec self-setup will not be used during X2 self-setup. After the X2 control plane is set up, coordination services will trigger the X2 user-plane setup and direct IPsec tunnel setup procedures between local and peer base stations through message exchange over the X2 control plane. However, the setup procedures will not be triggered if a user-plane self-setup to the peer base station has been performed within 10 minutes after the X2 control-plane self-setup is triggered.

NOTE

- After direct IPv6 IPsec is enabled, the control- or user-plane address of the X2 interface cannot be accessed directly through external networks. That is, the X2 interface cannot be negotiated as a plaintext channel.
- Direct IPv6 IPsec self-setup requires that the routing domains of the X2 control planes and user planes be the same. That is, the VRF indexes in the **SCTPHOST** and **USERPLANEHOST** MOs of all base stations enabled with direct IPv6 IPsec self-setup must be the same.

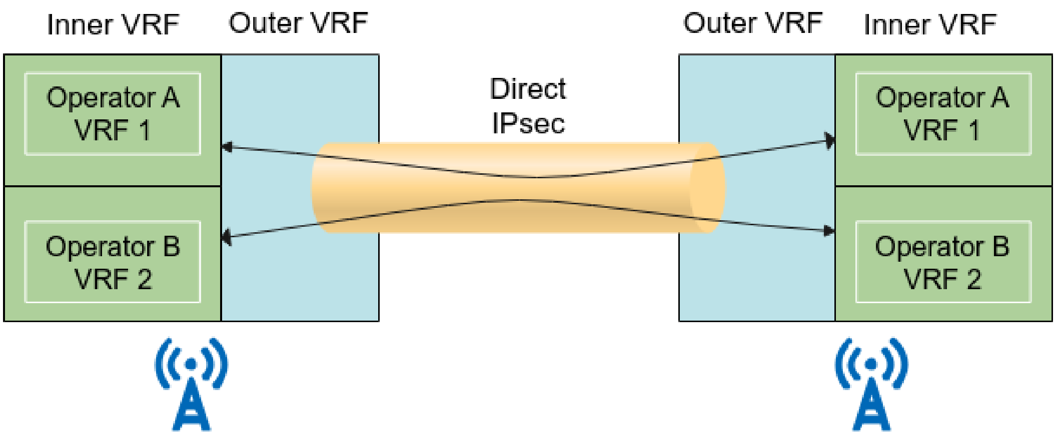
When both a direct IPv6 IPsec tunnel and an IPv6 IPsec tunnel passing through the SeGW are configured over the X2 interface, the direct IPv6 IPsec tunnel is preferentially used. This is because the **POLICYBASEDROUTING6** MO has a higher priority than the **IPROUTE6** and **SRCIPROUTE6** MOs.

Assume that the direct IPsec or **SECURITYPEER** specifications cannot meet the service requirements. When the **GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **STRICT_MODE**, a manually configured non-direct-IPsec X2 interface will be removed due to insufficient direct IPv6 IPsec specifications when the **SCTPHOST** or **USERPLANEHOST** changes from non-direct-IPv6-IPsec mode to direct-IPv6-IPsec mode. When the **GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **LOOSE_MODE**, an X2 interface is automatically generated even if direct IPv6 IPsec setup fails.

6.1.1.3 Direct IPsec Sharing Among Multiple Operators

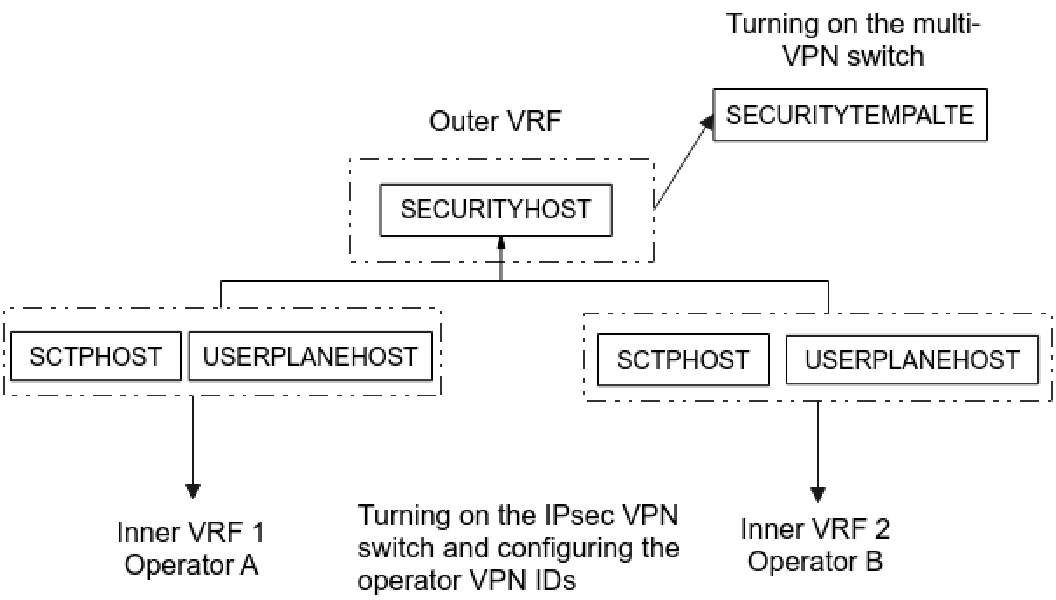
In multi-operator sharing scenarios, a direct IPsec tunnel can now be shared among operators over the X2 interface using the direct IPsec demultiplexing technology. Without the need for additional direct IPsec tunnels, direct IPsec services can be used by multiple operators at the same time based on their virtual private network (VPN) IDs. This saves resources and improves the usage rate of direct IPsec. Different inner VRFs are configured for different operators. IPsec can use VRFs to isolate outer packets from inner service packets. This function enables deployment of multiple operators on different inner VRFs of a base station. In this way, multiple operators can use a direct IPsec tunnel to share X2 services. In addition, X2 services no longer need to pass through the SeGW. **Figure 6-4** shows the principle using two operators as an example.

Figure 6-4 Direct IPsec sharing among multiple operators



In multi-operator sharing scenarios, after the multi-VPN switch is turned on (by using the **SECURITYTEMPLATE**.***MULTIVPNSWITCH (LTE eNodeB, 5G gNodeB)*** parameter), the IPsec VPN switch is turned on (by using the **VRF**.***IPSECVPNSW (LTE eNodeB, 5G gNodeB)*** parameter) and the corresponding operator VPN ID is configured (by using the **VRF**.***IPSECVPNID (LTE eNodeB, 5G gNodeB)*** parameter) to implement direct IPsec sharing among multiple operators. The VPN IDs of the operators must be unique and consistent on all the base stations involved. Otherwise, features or functions cannot work properly. **Figure 6-5** shows the configuration model. After the multi-VPN switch is turned on, multiple operators (up to six) can reference the same direct IPsec tunnel. The multi-VPN switch referenced by the **SECURITYTEMPLATE** MO and the IPsec VPN switch referenced by the **SCTPHOST** and **USERPLANEHOST** MOs must be both turned on or off.

Figure 6-5 Configuration model of direct IPsec sharing among multiple operators



6.1.2 Static Blacklist & Whitelist

Static Blacklist

The **ADD DIRECTIPSECBLACKLIST** command can be executed on a gNodeB or an eNodeB to configure a static direct IPsec blacklist.

If a security peer is included in the static direct IPsec blacklist of a local base station, the local base station does not establish a direct IPsec tunnel to the peer.

You are advised to configure the static direct IPsec blacklist before enabling the direct IPsec function. If direct IPsec tunnels have been set up between the local and peer base stations before the blacklist is configured, you need to manually remove the **SECURITYPEER** MO before adding a static direct IPsec blacklist.

Static Whitelist

A local base station will establish a direct IPsec tunnel to the peer in a network segment that is included in the static direct IPsec whitelist of the local base station.

If a gNodeB only needs to establish direct IPsec tunnels to peers in some network segments, you can add these network segments to the static direct IPsec whitelist.

The **ADD DIRECTIPSECWHITELIST** command can be executed on a gNodeB to configure a static direct IPsec whitelist.

NOTE

- The static blacklist function is mutually exclusive with the static whitelist function.
- The static whitelist function is mutually exclusive with the direct IPsec self-removal function.

6.1.3 Self-Update

gNodeBs/eNodeBs support the direct IPsec self-update function. This function is controlled by the **GEPMODELPARA**.***DIRIPSECUPDATESW (5G gNodeB, LTE eNodeB)*** parameter.

When this function is enabled and X2 interfaces are functioning, an X2 self-setup procedure is triggered for each X2 interface of a local base station if the X2 control plane is still functioning after any of the configuration modifications below is performed. Then, the local base station sends information about direct IPsec to the peer base stations. Direct IPsec self-update is performed if direct IPsec is enabled on the peer base stations.

- The value of the **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameter corresponding to the **EPGROUP** MO of an X2 interface is changed from **DISABLE** to **ENABLE**, and the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter value is **DISABLE**.
- The value of the **SECURITYHOST**.***SECURITYHOSTID (5G gNodeB, LTE eNodeB)*** parameter corresponding to the endpoint group of an X2 interface is changed, and the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter value is **DISABLE**. This trigger condition applies only to the IPv4 networking.

- The value of the **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameter corresponding to the endpoint group of an X2 interface is **ENABLE**, and the value of the **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter is changed from **ENABLE** to **DISABLE**. This trigger condition applies only to the IPv4 networking.
- The value of the **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameter corresponding to the endpoint group of an X2 interface is **ENABLE**, the value of the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter is **DISABLE**, and the **SECURITYHOST**.***LOCIKEIPV4 (5G gNodeB, LTE eNodeB)*** parameter value is changed.
- The value of the **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameter corresponding to the endpoint group of an X2 interface is **ENABLE**, the value of the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter is **DISABLE**, and the **SECURITYHOST**.***LOCIKEIPV6 (5G gNodeB, LTE eNodeB)*** parameter value is changed.
- The **ADD X2** command is executed to add an X2 interface. This operation is supported only on the LTE side, and different control-plane links must be used between the two X2 interfaces.
- On the LTE side, the **MOD X2** command is executed to change the value of the **X2**.***UpEpGroupId*** parameter. On the NR side, the **MOD GNBCUX2** command is executed to change the value of the **GNBCUX2**.***UpEpGroupId*** parameter.
- The value of the **SCTPHOST**.***SIGIP1SECSWITCH (LTE eNodeB, 5G gNodeB)*** parameter corresponding to the **EPGROUP** MO of an X2 interface is changed from **DISABLE** to **ENABLE**, and the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter value is **DISABLE**.
- The value of the **SCTPHOST**.***SIGIP1SECSWITCH (LTE eNodeB, 5G gNodeB)*** parameter corresponding to the endpoint group of an X2 interface is **ENABLE**, and the value of the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter is changed from **ENABLE** to **DISABLE**.
- The value of the **SCTPHOST**.***SIGIP1SECSWITCH (LTE eNodeB, 5G gNodeB)*** parameter corresponding to the endpoint group of an X2 interface is **ENABLE**, the value of the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter is **DISABLE**, and the **SECURITYHOST**.***LOCIKEIPV4 (5G gNodeB, LTE eNodeB)*** parameter value is changed.
- The value of the **SCTPHOST**.***SIGIP1SECSWITCH (LTE eNodeB, 5G gNodeB)*** parameter corresponding to the endpoint group of an X2 interface is **ENABLE**, the value of the corresponding **SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)*** parameter is **DISABLE**, and the **SECURITYHOST**.***LOCIKEIPV6 (5G gNodeB, LTE eNodeB)*** parameter value is changed.

Direct IPsec self-update takes effect only when X2 interfaces are functioning properly. Self-update of direct IPsec does not apply to an operator in RAN sharing with common carriers scenarios if the following conditions are met: (1) No X2 control plane is configured for the operator; (2) The operator shares the X2 control plane with another operator; (3) The X2 user plane is not shared between the two operators.

There is a 3-minute configuration protection period for the self-update of direct IPsec. After any of the preceding configuration changes takes place, the gNodeB/eNodeB configuration protection is triggered. After the protection period expires, the gNodeB/eNodeB determines whether to trigger a self-update procedure of direct IPsec based on the modifications to configurations.

After an X2 interface between the base station and any neighboring base station is automatically set up, a 100s protection timer is automatically started. In direct IPsec scenarios, if a user-plane IP address is changed within the protection timer length, the update of the X2 interface fails. The X2 interface can be updated only after the protection timer expires and the X2 self-update procedure restarts.

At a 20-hour interval, the base station automatically checks for interfaces for which the switch specified by the **SCTPHOST**.***SIGIP1SECSWITCH (LTE eNodeB, 5G gNodeB)*** or **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** parameter is turned on but no direct IPsec tunnels are established. If the IPsec self-update switch is turned on for such an interface, the base station automatically triggers direct IPsec setup. When the transmission mode is changed from IPv4 plaintext transmission to direct IPsec transmission, the interface will be disconnected if IPsec setup fails. If the IPsec self-update switch is turned off, the base station does not automatically trigger direct IPsec setup.

NOTE

Self-update is not supported for the direct IPsec tunnel for the X2-U interface that is set up based on the automatic peer configuration between gNodeBs. If the security host IP address of a gNodeB is changed, the following operations are performed:

- If the X2 control plane is normal and direct IPsec is configured on the local end, direct IPsec self-update is initiated on the user plane in the case of a base station reset.
- If an X2-U link becomes faulty due to a direct IPsec fault and the GTP-U detection result indicates that the fault lasts for more than one hour, the base station automatically removes the faulty X2-U link and the faulty direct IPsec tunnel according to the X2-U aging mechanism. (The GTP-U detection is performed only when the switch is turned on.) For details, see **X2-U Peer Self-Removal**.

6.1.4 Self-Removal

Self-removal of direct IPsec can be used when direct IPsec is enabled on a gNodeB/eNodeB and its connected peer eNodeB/gNodeB, but direct IPsec tunnels cannot be set up between the base station pair due to transport network planning reasons. This function is not recommended in other scenarios. This function is controlled by the **GEPMODELPARA**.***DIRIPSECAUTODELSW (LTE eNodeB, 5G gNodeB)*** parameter.

Assume that the **GEPMODELPARA**.***DIRIPSECAUTODELTIMER*** parameter is set to a non-zero value. When a gNodeB/eNodeB detects a direct IPsec tunnel fault caused by IKE SA negotiation failure, the gNodeB/eNodeB reports ALM-25891 IKE Negotiation Failure to facilitate transmission fault locating. The gNodeB/eNodeB then waits for a period specified by the **GEPMODELPARA**.***DIRIPSECAUTODELTIMER*** parameter. If the fault persists after the timer expires, the gNodeB/eNodeB removes the direct IPsec tunnel and ALM-25891 IKE Negotiation Failure is cleared. It is recommended that the **GEPMODELPARA**.***DIRIPSECAUTODELTIMER*** parameter be set to a value indicating one week or longer. If it is set to a value too small, the following cases may arise: (1) The gNodeB/eNodeB has removed the direct IPsec tunnel and the alarm has been cleared before users troubleshoot transmission faults based on the alarm. (2) The detection mechanism causes ALM-25891 IKE Negotiation Failure to be reported again after being cleared.

If gNodeB/eNodeB configuration is prohibited, or if there are a large number of direct IPsec tunnels to be removed, the fault recovery time is longer than the preset self-removal time (specified by the **GEPMODELPARA**.***DIRIPSECAUTODELTIMER*** parameter). This gap depends on the gNodeB/eNodeB configuration prohibition duration or the number of direct IPsec tunnels to be removed.

6.1.5 Application Restrictions

Direct IPsec does not apply to any X2-U interface that uses multiple IP addresses of the same version.

When direct IPsec is used for the X2 interface, the **SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)*** and **USERPLANEHOST**.***IPSECSWITCH (LTE eNodeB, 5G gNodeB)*** parameters on the local base station must be set to the same values as those on the peer base station.

Direct IPv6 IPsec requires that the value of the **EPGROUP**.***IPV6VRFIDX (5G gNodeB, LTE eNodeB)*** parameter differ from that of the **SECURITYHOST**.***VRFIDX (5G gNodeB, LTE eNodeB)*** parameter.

In IPv4/IPv6 dual-stack networking where direct IPsec is used for both IPv4 and IPv6, the method of direct IPv4 IPsec self-setup is determined by the settings of the **USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)*** and **SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)*** parameters.

Direct IPsec self-setup does not apply to IPv4-over-IPv6 IPsec (IPv4 for the inner IP header and IPv6 for the outer IP header) or IPv6-over-IPv4 IPsec (IPv6 for the inner IP header and IPv4 for the outer IP header). For details about IPv4-over-IPv6 IPsec and IPv6-over-IPv4 IPsec, see *IPsec*.

If **GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)*** is set to **LOOSE_MODE**, direct IPsec self-setup can be enabled when the **SCTPPEER**/**USERPLANEPEER** MO is automatically configured; manual setup of direct IPsec is not supported when the **SCTPPEER**/**USERPLANEPEER** MOs are manually configured.

In loose mode of direct IPsec self-setup, when only one direct IPsec tunnel can be set up according to the specifications, non-direct IPsec tunnels must be set up for both the **SCTPPEER** and **USERPLANEPEER** MOs.

Modifying the **GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)*** configurations does not affect the X2 links that have been set up.

Direct IPsec requires normal and stable transmission between base stations. If the transmission status changes frequently, IKE and IPsec negotiations will be frequently performed between base stations. As a result, the CPU usage of the main control board (**VS.BBUBoard.CPULoad.Mean (LTE eNodeB, 5G gNodeB)** or **VS.BBUBoard.CPULoad.Max (LTE eNodeB, 5G gNodeB)**) increases.

6.2 Network Analysis

6.2.1 Benefits

On a network secured using IPsec through SeGW, there can be significant delay over the X2 interface. Direct IPsec for X2 interface shortens the transmission path and reduces the delay over the X2 interface. When a large amount of X2 service data is transmitted between the eNodeB and the gNodeB through direct IPsec tunnels, traffic specifications of the SeGW are not occupied. This reduces network construction costs for operators.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

The change in this feature has no impact.

6.3 Requirements

Direct IPsec for X2 interface is based on X2 self-management. In addition to those described in **4.7 Requirements**, this function has the requirements described in this section.

6.3.1 Licenses

The license control items described in the following table are required on the eNodeB side but are not required on the gNodeB side.

Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
LOFD-121213	Direct IPsec	LT1SDIIPSEC0	Direct IPsec	eNodeB	per eNodeB
TDLOFD-121302	Direct IPsec	LT4SDIPSEC00	Direct IPsec	eNodeB	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	IPsec	None	*IPsec*	None
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	IPsec for IPv6	None	*IPsec*	None

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Transmission Configuration Mode	**GTRANSPARA.*TRANSCFGMODE (LTE eNodeB, 5G gNodeB)*** set to **NEW**	*X2 and S1 Self-Management in NSA Networking*	The multi-VPN switch (specified by the **SECURITYTEMPLATE.*MULTIVPNSWITCH (LTE eNodeB, 5G gNodeB)*** parameter) can be turned on only when the new transmission configuration model is used.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Self-removal of direct IPsec	**GEPMODELPARA.*DIRIPSECAUTODELSW (LTE eNodeB, 5G gNodeB)***	*X2 and S1 Self-Management in NSA Networking*	The multi-VPN switch (specified by the **SECURITYTEMPLATE.*MULTIVPNSWITCH (LTE eNodeB, 5G gNodeB)*** parameter) can be turned on only when the switch for self-removal of direct IPsec is turned on.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Tunnel-interface-based IPsec	**IKECFG.*IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** set to **TUNNELITF_BASED**	*X2 and S1 Self-Management in NSA Networking*	The multi-VPN switch (specified by the **SECURITYTEMPLATE.*MULTIVPNSWITCH (LTE eNodeB, 5G gNodeB)*** parameter) can be turned on only when the tunnel-interface-based IPsec configuration mode is used.

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	IPsec configuration mode	**IKECFG.*IPSECCFGMODE (LTE eNodeB, 5G gNodeB)***	*X2 and S1 Self-Management in NSA Networking*	In IPv4 scenarios, if **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** is set to **ETHINTERFACE_BASED**, the multi-VPN switch (**SECURITYTEMPLATE**.***MULTIVPNSWITCH (LTE eNodeB, 5G gNodeB)***) of the **SECURITYTEMPLATE** MO referenced by the **SECURITYHOST** MO for direct IPsec (with **SECURITYHOST**.***SEGWSWITCH (LTE eNodeB, 5G gNodeB)*** set to **DISABLE**) cannot be turned on.

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	RAN sharing with dedicated carrier/RAN sharing with common carrier	**gNBSharingMode**.***gNBMultiOpSharingMode***	*Multi-Operator Sharing* in *5G RAN Feature Documentation*	Direct IPsec self-setup is incompatible with multi-operator sharing.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The "direct IPsec sharing by operators over the X2 interface" function requires UMPTe series and later main control boards.

Other functions

For LTE, a UMPT is configured as the main control board.

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

N/A

6.3.4 Networking

If service delay requirements are not met for X2 interfaces that use IPsec tunnels passing through the SeGW, it is recommended that the transmission delays of direct transmission paths of the X2 interface be measured before direct IPsec is deployed:

- If the measured delay is less than the transmission delay of indirect transmission paths of the X2 interface and meets the delay requirements for to-be-deployed coordination-based features, the transport network performance allows for direct IPsec deployment.
- Otherwise, the transport network performance does not allow for direct IPsec deployment. In this case, you are advised to optimize the transport network first and deploy direct IPsec after the deployment conditions are met.

6.3.5 Others

The base stations on both sides of the X2 interface must be Huawei base stations.

When direct IPsec is required on the X2 interface where the gNodeB and eNodeB at each end of the X2 interface are located in different network segments, each of the two base stations must be configured with a route to the peer end.

Direct IPsec does not apply to X2-C interfaces adopting SCTP dual-homing.

Direct IPsec does not apply to any X2-U interface that uses multiple IP addresses of the same version.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Direct IPsec for the X2 interface is based on X2 self-setup between an eNodeB and a gNodeB. For details about the configurations for X2 self-setup, see **4.8.1.1 Data Preparation**. This section describes only IPsec configurations.

The following table describes the parameters that must be set in a **GTRANSPARA** MO used to configure the direct IPsec priority matching switch.

Parameter Name	Parameter ID	Setting Notes
Direct IPSec Priorities Match Switch	**GTRANSPARA**.***DIRECTIPSECPRIMATCHSW (5G gNodeB, LTE eNodeB)***	The value **ENABLE** is recommended. This parameter applies only to direct IPv4 IPsec.

The following table describes the parameters that must be set in an **IKECFG** MO to configure the IPsec configuration mode.

Parameter Name	Parameter ID	Setting Notes
IPSec Config Mode	**IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)***	The value **TUNNELITF_BASED** is recommended. If the **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **TUNNELITF_BASED**, the **IKECFG**.***IPSECBINDMODE (LTE eNodeB, 5G gNodeB)*** parameter cannot be set to **ALL**.

The following table describes the parameters that must be set in an **IKEPROPOSAL** MO.

Parameter Name	Parameter ID	Setting Notes
Proposal ID	**IKEPROPOSAL**.***PROPID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Authentication Method	**IKEPROPOSAL**.***AUTHMETH (5G gNodeB, LTE eNodeB)***	Set this parameter to **IKE_CERT_SIG** when digital certificate authentication is used. Set this parameter to the same value for all base stations in the network.

The following table describes the parameters that must be set in an **IPSECPROPOSAL** MO.

Parameter Name	Parameter ID	Setting Notes
IPSec Proposal Name	**IPSECPROPOSAL**.***PROPNAME (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **SECURITYTEMPLATE** MO. This MO applies only to IPsec-enabled scenarios. When the following parameters are to be modified, the base station needs to adjust the negotiation status of all security links automatically created using the **SECURITYTEMPLATE** MO. The adjustment takes a long time. Therefore, you are advised to run the **MOD SECURITYTEMPLATE** command to modify the parameters, wait for about 30 seconds, and then run other commands.

Parameter Name	Parameter ID	Setting Notes
Security Parameters Template ID	**SECURITYTEMPLATE**.***SECURITYTEMPLATEID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
SeGW Switch	**SECURITYTEMPLATE**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)***	Set this parameter to **DISABLE** when direct IPsec is required.
IKE Proposal ID	**SECURITYTEMPLATE**.***PROPID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
IPSec Proposal Name	**SECURITYTEMPLATE**.***IPSECPROPNAME (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
IKE Version	**SECURITYTEMPLATE**.***IKEVERSION (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Certificate Source	**SECURITYTEMPLATE**.***CERTSOURCE (5G gNodeB, LTE eNodeB)***	This parameter specifies the source of the certificate used for IKE negotiation in the multi-PKI scenario. Set this parameter based on the network plan.
Certificate File Name	**SECURITYTEMPLATE**.***CERTNAME (5G gNodeB, LTE eNodeB)***	This parameter specifies the name of the certificate file used for IKE negotiation in the multi-PKI scenario. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
IKEv2 Fragmentation Switch	**SECURITYTEMPLATE.*IKEV2FRGSW (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the network plan. The default value is **OFF**. For details about IKEv2 packet fragmentation, see *IPsec*.
IKEv2 Fragmentation MTU	**SECURITYTEMPLATE.*IKEV2FRGMTU (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the network plan. For details about IKEv2 packet fragmentation, see *IPsec*.
Trust Group ID	**SECURITYTEMPLATE.*TRUSTGROUPID (LTE eNodeB, 5G gNodeB)***	This parameter specifies the ID of a trust group. When certificate-based authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the **TRUSTGROUP** MO are used to authenticate the peer end. If this parameter is set to **65535**, no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.
OCSP Switch	**SECURITYTEMPLATE.*OCSPSW (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the network plan. The default value is **OFF**. For details about IKEv2 OCSP principles, see *IPsec*.
CRL Using Policy	**SECURITYTEMPLATE.*CRLPOLICY (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the network plan. The default value is **GLOBAL**. For details about CRL using policies, see *IPsec*.

The following table describes the parameters that must be set in a **SECURITYHOST** MO.

Parameter Name	Parameter ID	Setting Notes
Security Host ID	**SECURITYHOST**.***SECURITYHOSTID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
SeGW Switch	**SECURITYHOST**.***SEGWSWITCH (5G gNodeB, LTE eNodeB)***	Set this parameter to **DISABLE** when direct IPsec is required.
IP Version	**SECURITYHOST**.***IPVERSION (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Local IKE IP Address	**SECURITYHOST**.***LOCIKEIPV4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Local IKE IPv6 Address	**SECURITYHOST**.***LOCIKEIPV6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Security Parameters Template ID	**SECURITYHOST**.***SECURITYTEMPLATEID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	**SECURITYHOST**.***VRFIDX (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan. ● In IPv6, the value of this parameter must be different from the values of the **SCTPHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** and **USERPLANEHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameters. ● In IPv4, if the **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **ETHINTERFACE_BASED**, the value of this parameter must be the same as the values of the **SCTPHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** and **USERPLANEHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameters. If the **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **TUNNELITF_BASED**, the value of this parameter must be different from the values of the **SCTPHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** and **USERPLANEHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameters.

The following table describes the parameters that must be set in an **SCTPHOST** MO to configure information about the local base station of the X2-C interface.

Parameter Name	Parameter ID	Setting Notes
SCTP Host ID	**SCTPHOST**.***SCTPHOSTID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	**SCTPHOST**.***VRFIDX (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan. ● In IPv6, the value of this parameter must be the same as that of the **USERPLANEHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter but different from that of the **SECURITYHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter. ● In IPv4, if the **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **ETHINTERFACE_BASED**, the value of this parameter must be the same as the values of the **USERPLANEHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** and **SECURITYHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameters. If the **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **TUNNELITF_BASED**, the value of this parameter must be the same as that of the **USERPLANEHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter but different from that of the **SECURITYHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter.

Parameter Name	Parameter ID	Setting Notes
IP Version	**SCTPHOST**.***IPVERSION (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
First Local IP Address	**SCTPHOST**.***SIGIP1V4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
First Local IPv6 Address	**SCTPHOST**.***SIGIP1V6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
IPSec Auto Configure Switch for First Local IP	**SCTPHOST**.***SIGIP1SECSWITCH (5G gNodeB, LTE eNodeB)***	Set this parameter to **ENABLE** when control-plane direct IPsec is required.
Security Host ID for First Local IP	**SCTPHOST**.***SIGIP1SECHOSTID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Local SCTP Port No.	**SCTPHOST**.***PN (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
SCTP Parameters Template ID	**SCTPHOST**.***SCTPTEMPLATEID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO to configure information about the local base station of the X2-U interface.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	**USERPLANEHOST**.***UPHOSTID (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	**USERPLANEHOST**.***VRFIDX (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan. ● In IPv6, the value of this parameter must be the same as that of the **SCTPHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter but different from that of the **SECURITYHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter. ● In IPv4, if the **IKECFG**.***IPSECCFGMODE (5G gNodeB, LTE eNodeB)*** parameter is set to **ETHINTERFACE_BASED**, the value of this parameter must be the same as the values of the **SCTPHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** and **SECURITYHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameters. If the **IKECFG**.***IPSECCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **TUNNELITF_BASED**, the value of this parameter must be the same as that of the **SCTPHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter but different from that of the **SECURITYHOST**.***VRFIDX (LTE eNodeB, 5G gNodeB)*** parameter.

Parameter Name	Parameter ID	Setting Notes
IP Version	**USERPLANEHOST**.***IPVERSION (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Local IP Address	**USERPLANEHOST**.***LOCIPV4 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Local IPv6 Address	**USERPLANEHOST**.***LOCIPV6 (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
IPSec Auto Configure Switch	**USERPLANEHOST**.***IPSECSWITCH (5G gNodeB, LTE eNodeB)***	Set this parameter to **ENABLE** when direct IPsec is required.
Security Host ID	**USERPLANEHOST**.***SECHOSTID (5G gNodeB, LTE eNodeB)***	Set this parameter to the same value as the security host ID in the **SCTPHOST** MO when direct IPsec is required.

(Optional) The following table describes the parameters that must be set in a **GEPMODELPARA** MO to configure direct IPsec self-removal.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Automatic Deletion Switch	**GEPMODELPARA**.***DIRIPSECAUTODELSW (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.
Direct IPsec Automatic Deletion Timer	**GEPMODELPARA**.***DIRIPSECAUTODELTIMER (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **GEPMODELPARA** MO to configure direct IPsec self-setup.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Automatic Setup Mode	**GEPMODELPARA**.***DIRECTIPSECAUTOSETUPMODE (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

(Optional) The following table describes the parameter that must be set in a **GEPMODELPARA** MO to configure direct IPsec self-update.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Update Switch	**GEPMODELPARA**.***DIRIPSECUPDATESW (5G gNodeB, LTE eNodeB)***	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **DIRECTIPSECBLACKLIST** MO to configure a direct IPsec blacklist.

Parameter Name	Parameter ID	Setting Notes
ID of Blacklist for Direct IPsec Tunnel	**DIRECTIPSECBLACKLIST**.***DIRECTIPSECBLACKLISTID***	Set this parameter based on the network plan.
IP Version	**DIRECTIPSECBLACKLIST**.***IPVERSION***	Set this parameter based on the network plan.
Security Peer IPv4 Address	**DIRECTIPSECBLACKLIST**.***SECURITYPEERIPV4***	Set this parameter based on the network plan. This parameter is valid only when the **DIRECTIPSECBLACKLIST**.***IPVERSION*** parameter is set to **IPV4**.
Security Peer IPv6 Address	**DIRECTIPSECBLACKLIST**.***SECURITYPEERIPV6***	Set this parameter based on the network plan. This parameter is valid only when the **DIRECTIPSECBLACKLIST**.***IPVERSION*** parameter is set to **IPV6**.

The following table describes the parameters that must be set in a **DIRECTIPSECWHITELIST** MO to configure a direct IPsec whitelist.

Parameter Name	Parameter ID	Setting Notes
ID of Whitelist for Direct IPsec Tunnel	**DIRECTIPSECWHITELIST**.***DIRECTIPSECWHITELISTID***	Set this parameter based on the network plan.
IP Version	**DIRECTIPSECWHITELIST**.***IPVERSION***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Security Peer IPv4 Address	**DIRECTIPSECWHITELIST**.***SECURITYPEERIPV4***	Set this parameter based on the network plan. This parameter is valid only when the **DIRECTIPSECWHITELIST**.***IPVERSION*** parameter is set to **IPV4**.
Mask	**DIRECTIPSECWHITELIST**.***MASK***	Set this parameter based on the network plan.
Security Peer IPv6 Address	**DIRECTIPSECWHITELIST**.***SECURITYPEERIPV6***	Set this parameter based on the network plan. This parameter is valid only when the **DIRECTIPSECWHITELIST**.***IPVERSION*** parameter is set to **IPV6**.
IPv6 Address Prefix Length	**DIRECTIPSECWHITELIST**.***IPV6PFXLEN***	Set this parameter based on the network plan. This parameter is valid only when the **DIRECTIPSECWHITELIST**.***IPVERSION*** parameter is set to **IPV6**.

(Optional) Configure the **VRF** MO to configure direct IPsec sharing among multiple operators over the X2 interface.

Parameter Name	Parameter ID	Setting Notes
Multiple VPN Switch	**SECURITYTEMPLATE**.***MULTIVPNSWITCH (5G gNodeB, LTE eNodeB)***	Set this parameter to **ON** when direct IPsec sharing among multiple operators over the X2 interface is enabled.
IPSec VPN Switch	**VRF**.***IPSECVPNSW (LTE eNodeB, 5G gNodeB)***	Set this parameter to **ENABLE** when direct IPsec sharing among multiple operators over the X2 interface is enabled.

Parameter Name	Parameter ID	Setting Notes
IPSec VPN ID	**VRF**.***IPSECVPNID (LTE eNodeB, 5G gNodeB)***	Set this parameter based on the network plan. Operators must negotiate with each other and ensure that the parameter setting of each operator is consistent across all of its base stations.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activating Direct IPsec for the X2 Interface

Direct IPv4 IPsec self-setup:

- Scenario 1: The IPv4 IPsec configuration mode is Ethernet-interface-based IPsec.

  Configuring parameters on the eNodeB

```
//Enabling X2 self-setup and specifying the X2 self-setup mode, with direct IPsec set for the user plane as an example and coordination-based services in non-ideal backhaul mode carried by the X2 interface
MOD GLOBALPROCSWITCH: X2SonSetupSwitch=ON, X2SonLinkSetupType=X2_OVER_S1, ItfTypeForNonIdealModeServ=X2;
//Setting the direct IPsec matching priority with Direct IPSec Priorities Match Switch set to ENABLE (This setting takes effect only for IPv4 IPsec.)
SET GTRANSPARA: DIRECTIPSECPRIMATCHSW=ENABLE;
//Setting IPsec Config Mode to ETHINTERFACE_BASED
SET IKECFG: IPSECCFGMODE=ETHINTERFACE_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=DES, AUTHALG=MD5, DURATION=86400, REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP, ESPAUTHALG=MD5, ESPENCALG=DES;
//Adding a security parameter template for direct IPv4 IPsec self-setup with SeGW Switch set to DISABLE
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10, IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="******", DPD=PERIODIC, LTCFG=LOCAL, CERTSOURCE=APPCERT, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4, LOCIKEIPV4="192.168.4.146", SECURITYTEMPLATEID=0;
//Adding an endpoint group for the X2 interface
```

```
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1, IPPMSWITCH=DISABLE, APPTYPE=NULL;
//Adding an SCTP parameter template
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an X2 control-plane host
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="192.168.42.93",
SIGIP1SECSWITCH=DISABLE, SIGIP2V4="192.168.0.0", SIGIP2SECSWITCH=DISABLE, PN=36422,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0;
//Adding an X2 user-plane host
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="192.168.42.93",
IPSECSWITCH=ENABLE, SECHOSTID=0;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//(Optional) Adding a transmission resource group. This command is required only when
GTRANSPARA.TRANSCFGMODE is set to OLD and a dedicated transmission resource group needs to
be specified for the X2 interface.
ADD RSCGRP: CN=0, SRN=1, SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0, RU=KBPS;
//(Optional) Adding the X2 user plane to the transmission resource group. This command is required
only when GTRANSPARA.TRANSCFGMODE is set to OLD and a dedicated transmission resource group
needs to be specified for the X2 interface.
ADD EP2RSCGRP: ENDPOINTID=1, CN=0, SRN=1, SN=7, MT=ENDPOINT_GROUP, SBT=BASE_BOARD,
PT=ETH, BEAR=IP, RSCGRPID=0;
//(Optional) Adding an IP transmission resource group. This command is required only when
GTRANSPARA.TRANSCFGMODE is set to NEW and a dedicated IP transmission resource group needs
to be specified for the X2 interface.
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=0, RU=KBPS;
//(Optional) Adding the mapping relationship between the endpoint group and IP transmission
resource group. This command is required only when GTRANSPARA.TRANSCFGMODE is set to NEW
and a dedicated IP transmission resource group needs to be specified for the X2 interface.
ADD EP2IPRSCGRP: MAPID=2, MT=ENDPOINT_GROUP, ENDPOINTID=1, IPRSCGRPID=0;
//Enabling ANR
MOD ENODEBALGOSWITCH: AnrSwitch=IntraRatEventAnrSwitch-1&IntraRatFastAnrSwitch-1;
MOD CELLALGOSWITCH:LocalCellId=0,AnrFunctionSwitch=INTRA_RAT_ANR_SW-1;
//Adding an X2 object
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
```

Configuring parameters on the gNodeB

```
//Enabling X2 self-setup and specifying the X2 self-setup mode, with direct IPsec set for the user
plane as an example and coordination-based services in non-ideal backhaul mode carried by the X2
interface
MOD GLOBALPROCSWITCH: X2SonSetupSwitch=ON, X2SonLinkSetupType=X2_OVER_S1,
ItfTypeForNonIdealModeServ=X2;
//Setting the direct IPsec matching priority with Direct IPSec Priorities Match Switch set to ENABLE
(This setting takes effect only for IPv4 IPsec.)
SET GTRANSPARA: DIRECTIPSECPRIMATCHSW=ENABLE;
//Setting IPsec Config Mode to ETHINTERFACE_BASED
SET IKECFG: IPSECCFGMODE=ETHINTERFACE_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=DES, AUTHALG=MD5, DURATION=86400, REAUTHSW=ON,
REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPAUTHALG=MD5, ESPENCALG=DES;
//Adding a security parameter template for direct IPv4 IPsec self-setup with SeGW Switch set to
DISABLE
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="******", DPD=PERIODIC, LTCFG=LOCAL,
CERTSOURCE=APPCERT, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4,
LOCIKEIPV4="192.168.4.146", SECURITYTEMPLATEID=0;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1, IPPMSWITCH=DISABLE, APPTYPE=NULL;
//Adding an SCTP parameter template
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an X2 control-plane host
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="192.168.42.93",
SIGIP1SECSWITCH=DISABLE, SIGIP2V4="192.168.0.0", SIGIP2SECSWITCH=DISABLE, PN=36422,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0;
```

```
//Adding an X2 user-plane host
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="192.168.42.93",
IPSECSWITCH=ENABLE, SECHOSTID=0;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//(Optional) Adding a transmission resource group. This command is required only when
GTRANSPARA.TRANSCFGMODE is set to OLD and a dedicated transmission resource group needs to
be specified for the X2 interface.
ADD RSCGRP: CN=0, SRN=1, SN=7, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=0, RU=KBPS;
//(Optional) Adding the X2 user plane to the transmission resource group. This command is required
only when GTRANSPARA.TRANSCFGMODE is set to OLD and a dedicated transmission resource group
needs to be specified for the X2 interface.
ADD EP2RSCGRP: ENDPOINTID=1, CN=0, SRN=1, SN=7, MT=ENDPOINT_GROUP, SBT=BASE_BOARD,
PT=ETH, BEAR=IP, RSCGRPID=0;
//(Optional) Adding an IP transmission resource group. This command is required only when
GTRANSPARA.TRANSCFGMODE is set to NEW and a dedicated IP transmission resource group needs
to be specified for the X2 interface.
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=0, RU=KBPS;
//(Optional) Adding the mapping relationship between the endpoint group and IP transmission
resource group. This command is required only when GTRANSPARA.TRANSCFGMODE is set to NEW
and a dedicated IP transmission resource group needs to be specified for the X2 interface.
ADD EP2IPRSCGRP: MAPID=2, MT=ENDPOINT_GROUP, ENDPOINTID=1, IPRSCGRPID=0;
//Enabling ANR
MOD ENODEBALGOSWITCH: AnrSwitch=IntraRatEventAnrSwitch-1&IntraRatFastAnrSwitch-1;
MOD CELLALGOSWITCH:LocalCellId=0,AnrFunctionSwitch=INTRA_RAT_ANR_SW-1;
//Adding an X2 object
ADD GNBCUX2: gNBCuX2Id=0, CpEpGroupId=1, UpEpGroupId=1;
```

- Scenario 2: The IPv4 IPsec configuration mode is tunnel-interface-based IPsec.

  Configuring parameters on the eNodeB

```
//Enabling X2 self-setup and specifying the X2 self-setup mode, with direct IPsec set for the user
plane as an example and coordination-based services in non-ideal backhaul mode carried by the X2
interface
MOD GLOBALPROCSWITCH: X2SonSetupSwitch=ON, X2SonLinkSetupType=X2_OVER_S1,
ItfTypeForNonIdealModeServ=X2;
//Setting the direct IPsec matching priority with Direct IPSec Priorities Match Switch set to ENABLE
(This setting takes effect only for IPv4 IPsec.)
SET GTRANSPARA: DIRECTIPSECPRIMATCHSW=ENABLE;
//Setting IPsec Config Mode to TUNNELITF_BASED
SET IKECFG: IPSECCFGMODE=TUNNELITF_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=DES, AUTHALG=MD5, DURATION=86400, REAUTHSW=ON,
REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPAUTHALG=MD5, ESPENCALG=DES;
//Adding a security parameter template to be used by the eNodeB for direct IPv4 IPsec self-setup,
with SeGW Switch set to DISABLE and Narrow Down Switch set to ON
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="******", DPD=PERIODIC,
CERTSOURCE=APPCERT, LTCFG=LOCAL, NARROWDOWNSW=ON, IKEV2FRGSW=OFF, OCSPSW=OFF,
CRLPOLICY=GLOBAL;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=2,
LOCIKEIPV4="192.168.4.146", SECURITYTEMPLATEID=0;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, VRFIDX=1, IPPMSWITCH=DISABLE, APPTYPE=NULL;
//Adding an SCTP parameter template
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an X2 control-plane host
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="192.168.42.93",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=0, SIGIP2SECSWITCH=DISABLE, PN=36422,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL;
//Adding an X2 user-plane host
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="192.168.42.93",
IPSECSWITCH=ENABLE, SECHOSTID=0;
//Adding the X2 user-plane host to the endpoint group
```

```
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//(Optional) Adding an IP transmission resource group. This command is required only when a
dedicated IP transmission resource group needs to be specified for the X2 interface.
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=0, RU=KBPS;
//(Optional) Adding the mapping relationship between the endpoint group and IP transmission
resource group. This command is required only when a dedicated IP transmission resource group
needs to be specified for the X2 interface.
ADD EP2IPRSCGRP: MAPID=2, MT=ENDPOINT_GROUP, ENDPOINTID=1, IPRSCGRPID=0;
//Enabling ANR
MOD ENODEBALGOSWITCH: AnrSwitch=IntraRatEventAnrSwitch-1&IntraRatFastAnrSwitch-1;
MOD CELLALGOSWITCH:LocalCellId=0,AnrFunctionSwitch=INTRA_RAT_ANR_SW-1;
//Adding an X2 object
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
```

Configuring parameters on the gNodeB

```
//Turning on the X2 self-setup switch
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_SETUP_SWITCH-1;
//Setting the direct IPsec matching priority with Direct IPSec Priorities Match Switch set to ENABLE
(This setting takes effect only for IPv4 IPsec.)
SET GTRANSPARA: DIRECTIPSECPRIMATCHSW=ENABLE;
//Setting IPsec Config Mode to TUNNELITF_BASED
SET IKECFG: IPSECCFGMODE=TUNNELITF_BASED;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHALG=SHA256, DURATION=86400,
REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPAUTHALG=SHA256, ESPENCALG=AES128;
//Adding a security parameter template to be used by the gNodeB for direct IPv4 IPsec self-setup,
with SeGW Switch set to DISABLE and Narrow Down Switch set to ON
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="******", DPD=PERIODIC,
CERTSOURCE=APPCERT, LTCFG=LOCAL, NARROWDOWNSW=ON, IKEV2FRGSW=OFF;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=2,
LOCIKEIPV4="7.7.4.146", SECURITYTEMPLATEID=0;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, VRFIDX=1;
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=1, SWITCHBACKFLAG=ENABLE;
//Adding an X2 control-plane host
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="7.7.42.93",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=0, SIGIP2SECSWITCH=DISABLE, PN=38422,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1, DTLSPOLICYID=NULL;
//Adding an X2 user-plane host
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="7.7.42.93",
IPSECSWITCH=ENABLE, SECHOSTID=0;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Enabling ANR
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch = NR_NR_ANR_SW-1&NR_EUTRAN_ANR_SW-1;
//Adding an X2 object
ADD GNBCUX2: gNBCuX2Id=0, CpEpGroupId=1, UpEpGroupId=1;
```

Direct IPv6 IPsec self-setup:

Configuring parameters on the eNodeB

```
//Turning on the X2 self-setup switch
MOD GLOBALPROCSWITCH: X2SonSetupSwitch=ON, InterfaceSetupPolicySw=LTE_NR_X2_SON_SETUP_SW-1;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES256, AUTHALG=SHA256, DURATION=86400,
REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
```

```
ESPAUTHALG=SHA256, ESPENCALG=AES256;
//Adding a security parameter template used when direct IPv6 IPsec tunnels are automatically set up by the base station
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=10, IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="******", DPD=PERIODIC, LTCFG=LOCAL, CERTSOURCE=APPCERT, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding a security host
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv6, LOCIKEIPV6="xx::xx:xx:xx", SECURITYTEMPLATEID=1, VRFIDX=2;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1, IPPMSWITCH=DISABLE, APPTYPE=NULL;
//Adding an SCTP parameter template
ADD SCTPTEMPLATE: SCTPTEMPLATEID=1, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an X2 control-plane host
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2V6="0::0", SIGIP2SECSWITCH=ENABLE, PN=36422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1;
//Adding an X2 user-plane host
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=ENABLE, SECHOSTID=1;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding an X2 object
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
```

Configuring parameters on the gNodeB

```
//Turning on the X2 self-setup switch
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_SETUP_SWITCH-1;
//Adding an IKE proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES256, AUTHALG=SHA256, DURATION=86400, REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP, ESPAUTHALG=SHA256, ESPENCALG=AES256;
//Adding a security parameter template used when direct IPv6 IPsec tunnels are automatically set up by the base station
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=10, IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="******", DPD=PERIODIC, LTCFG=LOCAL, CERTSOURCE=APPCERT, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding a security host
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv6, VRFIDX=2, LOCIKEIPV6="xx::xx:xx:xx", SECURITYTEMPLATEID=1;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1, IPPMSWITCH=DISABLE, APPTYPE=NULL;
//Adding an SCTP parameter template
ADD SCTPTEMPLATE: SCTPTEMPLATEID=1, SWITCHBACKFLAG=ENABLE;
//Adding an X2 control-plane host
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=ENABLE, SIGIP2V6="0:0:0:0:0:0:0:0", SIGIP2SECSWITCH=DISABLE, PN=36422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1, DTLSPOLICYID=NULL;
//Adding an X2 user-plane host
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=ENABLE, SECHOSTID=1;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding an X2 object
ADD GNBCUX2: gNBCuX2Id=0, CpEpGroupId=1, UpEpGroupId=1;
```

Activating Self-Update of Direct IPsec for the X2 Interface

```
SET GEPMODELPARA: DIRIPSECUPDATESW=ENABLE;
```

Activating Self-Removal of Direct IPsec for the X2 Interface

```
//Enabling self-removal of direct IPsec with the direct IPsec self-removal timer length being 10 minutes
SET GEPMODELPARA: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
```

Configuring the Mode of Direct IPsec Self-Setup for the X2 Interface

```
//Setting the direct IPsec self-setup mode to STRICT_MODE
SET GEPMODELPARA: DIRECTIPSECAUTOSETUPMODE=STRICT_MODE;
```

Configuring a Direct IPsec Blacklist for the X2 Interface

```
//Adding a direct IPsec blacklist entry
ADD DIRECTIPSECBLACKLIST: DIRECTIPSECBLACKLISTID=0, IPVERSION=IPV4, SECURITYPEERIPV4="X.X.X.X";
```

Configuring a Direct IPsec Whitelist for the X2 Interface

```
//Adding a direct IPsec whitelist entry
ADD DIRECTIPSECWHITELIST: DIRECTIPSECWHITELISTID=0, IPVERSION=IPV4, SECURITYPEERIPV4="x.x.x.x",
MASK="x.x.x.x";
```

Configuring Direct IPsec Sharing by Operators over the X2 Interface (Using Two Operators as an Example)

Configuration for a newly-deployed base station

```
//Configuring inner VRFs for the two operators
ADD VRF: VRFIDX=1, USERLABEL="OP1", IPSECVPNSW=ENABLE, IPSECVPNID=1;
ADD VRF: VRFIDX=2, USERLABEL="OP2", IPSECVPNSW=ENABLE, IPSECVPNID=2;
//Creating a security parameter template and enabling the multi-VPN function
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
ADD IPSECPROPOSAL: PROPNAME="1", TRANMODE=ESP, ESPENCALG=AES128;
SET GEPMODELPARA: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=1,
IPSECPROPNAME="1", IKEVERSION=IKE_V2, DPD=PERIODIC, CERTSOURCE=APPCERT, LTCFG=LOCAL,
IKEV2FRGSW=OFF, NARROWDOWNSW=ON, MULTIVPNSWITCH=ON;
//Configuring the security host information and referencing the preceding security parameter template
(VRF 0 is the outer VRF, and the IPsec VPN switch does not need to be turned on.)
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=0,
LOCIKEIPV4="10.1.1.1", SECURITYTEMPLATEID=1;
//Adding control-plane hosts for the two operators
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="1.1.1.1",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL, USERLABEL="OP1";
ADD SCTPHOST: SCTPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, SIGIP1V4="2.2.2.2",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL, USERLABEL="OP2";
//Adding user-plane hosts for the two operators
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="1.1.1.1",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP1";
ADD USERPLANEHOST: UPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, LOCIPV4="2.2.2.2",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP2";
```

Configurations in scenarios where operator B is added

```
//Modifying the control-plane host for operator A and disabling direct IPsec
MOD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="1.1.1.1",
SIGIP1SECSWITCH=DISABLE, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL, USERLABEL="OP1";
//Modifying the user-plane host for operator A and disabling direct IPsec
MOD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="1.1.1.1",
IPSECSWITCH=DISABLE, USERLABEL="OP1";
//Creating a security parameter template and enabling the multi-VPN function
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
ADD IPSECPROPOSAL: PROPNAME="1", TRANMODE=ESP, ESPENCALG=AES128;
SET GEPMODELPARA: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
```

```
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=1,
IPSECPROPNAME="1", IKEVERSION=IKE_V2, DPD=PERIODIC, CERTSOURCE=APPCERT, LTCFG=LOCAL,
IKEV2FRGSW=OFF, NARROWDOWNSW=ON, MULTIVPNSWITCH=ON;
//Configuring the security host information and referencing the preceding security parameter template
(VRF 0 is the outer VRF, and the IPsec VPN switch does not need to be turned on.)
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=0,
LOCIKEIPV4="10.1.1.1", SECURITYTEMPLATEID=1;
//Configuring inner VRFs for the two operators
MOD VRF: VRFIDX=1, USERLABEL="OP1", IPSECVPNSW=ENABLE, IPSECVPNID=1;
ADD VRF: VRFIDX=2, USERLABEL="OP2", IPSECVPNSW=ENABLE, IPSECVPNID=2;
//Modifying the control-plane host for operator A and enabling direct IPsec again
MOD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="1.1.1.1",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL, USERLABEL="OP1";
//Modifying the user-plane host for operator A and enabling direct IPsec again
MOD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="1.1.1.1",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP1";
//Adding a control-plane host for operator B
ADD SCTPHOST: SCTPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, SIGIP1V4="2.2.2.2",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL, USERLABEL="OP2";
//Adding a user-plane host for operator B
ADD USERPLANEHOST: UPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, LOCIPV4="2.2.2.2",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP2";
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Setting IPSec Auto Configure Switch for First Local IP and IPSec Auto Configure Switch for Second Local IP
of the SCTPHOST MO to DISABLE
MOD SCTPHOST: SCTPHOSTID=1, SIGIP1SECSWITCH=DISABLE, SIGIP2SECSWITCH=DISABLE;
//Setting IPSec Auto Configure Switch of the USERPLANEHOST MO to DISABLE
MOD USERPLANEHOST: UPHOSTID=1, IPSECSWITCH=DISABLE;
//Disabling direct IPsec sharing by operators over the X2 interface (using two operators as an example)
MOD VRF: VRFIDX=1, USERLABEL="OP1", IPSECVPNSW=DISABLE;
MOD VRF: VRFIDX=2, USERLABEL="OP2", IPSECVPNSW=DISABLE;
MOD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, IKEVERSION=IKE_V2,
MULTIVPNSWITCH=ON;
```

6.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Direct IPsec for the X2-U Interface (Query Through MML Commands)

Step 1 Run the **DSP IKESA** command to check the IKE SA negotiation results.

Expected result: The IDs of ACL rules generated during direct IPsec self-setup for the X2 interface range from 80000 to 84999. If the command output includes an IKE SA with the rule ID in this range, direct IPsec negotiation between the local and peer base stations has succeeded, and the state of the IKE SA between the base station pair is normal.

Step 2 Run the **LST IPSECPOLICY** command to obtain **Policy Group Name** and **IPSec Sequence No.** corresponding to the ACL rule ID queried in the previous step.

Step 3 Run the **DSP IPSECSA** command to display the results of IPsec SA negotiation based on **IPSec Policy Group Name** and **IPSec Sequence No.**

Expected result: IPsec SA status information is displayed, which indicates that direct IPsec negotiation between the local and peer base stations has succeeded, and IPsec SA between the base station pair is normal.

NOTE

This procedure uses direct IPsec SA setup for the X2-U interface as an example. If direct IPsec is successfully set up for the control plane between the local and the peer base stations, IPsec SA information about the control plane is also displayed.

----End

Direct IPsec for the X2-U Interface (Track Through Signaling Trace)

Step 1 Enable GTP-U static detection on the local and peer base stations by setting **GTPU**.***STATICCHK (LTE eNodeB, 5G gNodeB)*** to **ENABLE**. Enable periodical transmission of GTP-U packets to simulate user-plane packets.

Step 2 Log in to the MAE-Access. Choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** window is displayed.

Step 3 Choose **Trace Type** > **Base Station Device and Transport** > **Transport Trace** > **IP Layer Protocol Trace**.

Step 4 Set the protocol type to **GTPU** to start the IP layer protocol trace between the local and peer base stations and collect GTP-U packets.

Step 5 Parse the collected GTP-U packets to check whether user-plane packets are encrypted by direct IPsec.

----End

Self-Update of Direct IPsec

Wait 3 minutes after direct IPsec self-update is enabled. Then check the HUAWEI_PRIVATE_MSG messages traced over the X2 interface. If **From Peer Base Station** and **To Peer Base Station** are available in the **Trace Direction** column on the LTE side (as shown in **Figure 6-6**), and **To eNodeB** and **From eNodeB** are available in the **Trace Direction** column on the NR side (as shown in **Figure 6-7**), this function has been successfully activated.

Figure 6-6 X2 message tracing for self-update of direct IPsec on the LTE side

X2 Interface Trace(2097155)-[Direction:Bidirectional Trace object:All X2 Interface UE Type:MBB UE]

No.	Time	Standard Interface Mes...	Trace Direction	Peer NE ID
1	2018-06-26 18:54:13(169)	HUAWEI_PRIVATE_MSG	From Peer Base Station	722025
2	2018-06-26 18:54:13(171)	HUAWEI_PRIVATE_MSG	To Peer Base Station	722025

Figure 6-7 X2 message tracing for self-update of direct IPsec on the NR side

X2 Interface Trace(2097154)-[Trace Direction:Bidirectional Trace Object:All eNodeBs]

No.	Time	Standard Interface Me...	Trace Direction	X2 Interface ID	eNodeB ID
1	2018-06-26 18:54:13(689)	HUAWEI_PRIVATE_MSG	To eNodeB	2	722027
2	2018-06-26 18:54:13(696)	HUAWEI_PRIVATE_MSG	From eNodeB	2	722027

Self-Removal of Direct IPsec

If ALM-25891 IKE Negotiation Failure is reported, the alarm automatically clears when the **Direct IPsec Automatic Deletion Timer** expires (if this timer length is not 0). Then, run the **LST IKEPEER** command to view the configurations of the IKE peer.

Expected result: The peer IP address of the IKE peer is unavailable. This indicates that this function has taken effect.

6.4.3 Network Monitoring

None

7 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 37.340 "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity"
- 3GPP TS 36.423 "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 application protocol (X2AP)"
- RFC 7383 "Internet Key Exchange Protocol Version 2 (IKEv2) Message Fragmentation"
- *EPC-based NSA Basics*
- *NSA Mobility Management*
- *IPsec*
- *S1 and X2 Self-Management* in *eRAN Feature Documentation*
- *RAN Sharing* in *eRAN Feature Documentation*
- *Equipment Security*